Route Anything: A Theory of Inference Routers

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Abstract

An inference router chooses which model, tool, or combination should handle a task, using current state and past outcomes. We develop a theory organized around three router decisions: which capabilities to combine, when to pay for a better routing decision, and what information to retain. A reference-relative surplus identity compares routing policies while accounting for delivered quality, consumed resources, and routing overhead. Verified execution with escalation provides a concrete setting in which to derive decision criteria. For portfolio routers, an exact residual-coverage rule and a two-world lower bound show why individual model scores cannot identify useful complementarity. A value-of-information analysis determines when a stronger routing signal justifies its cost. A router-memory model with informative writes, erroneous writes, and expiration yields exact transient and stationary gains for finite and heavy-tailed task spaces. Reproducible analytical examples connect these results to the design of inference routers.

1. Introduction

An inference router chooses the next computation from a heterogeneous capability pool. A smaller model may solve an easy query, a specialist may resolve a particular uncertainty, and a tool may certify an answer that would otherwise require another model call. A router can select one model, combine complementary proposers with an aggregator, or assign different roles and context views. Its decision can occur once per task or repeatedly within an agent's execution. The central problem is to choose sufficient capability at a resource cost justified by the value it delivers.

Model routing and cascades already formalize important parts of this problem. FrugalGPT learns economical cascades (Chen et al., 2024), RouteLLM learns routing from preferences (Ong et al., 2025), and Dekoninck et al. (2025) unify routing and cascading under a budget. These settings distinguish a router's decision from its execution: an ex-ante router dispatches a selected path, whereas a verified cascade attempts a cheap path and escalates after observing its insufficiency. A portfolio router additionally decides which capabilities should work together. The value of these decisions depends on joint failures, routing overhead, and what the router has learned from earlier outcomes.

Speculative decoding offers a useful way to analyze verified execution with escalation. A fast draft proposes tokens and a target model verifies them while preserving the target distribution (Leviathan et al., 2023; Chen et al., 2023). Speculative speculative decoding (SSD) additionally predicts verification outcomes to prepare the next draft in parallel (Kumar et al., 2026). Its analysis organizes the design around useful speculation, allocation of speculative work, and fallback on a miss. For a router choosing complete model or tool calls, however, verification need not preserve value or have fixed cost. An unsuccessful attempt consumes resources before escalation, and changing a portfolio changes the tasks left to the fallback. A larger acceptance rate can therefore accompany lower value per resource.

We study *inference routers* as stateful decision policies. The router chooses a capability allocation and an execution rule; observed outcomes update the information available for later choices. Verified execution with escalation, which we call speculative capability allocation (SCA), is our principal setting for deriving explicit criteria. Our central question is:

> What should a router combine, when should it spend more, and what should it remember?

This question leads to three coupled router design problems. First, a portfolio router must identify capability that covers the current portfolio's failures. Second, a router must decide whether a cheap attempt or an additional routing signal is worth paying for. Third, router memory must retain information that improves subsequent decisions. All three require comparing the value of avoided fallback work with routing overhead and any change in delivered quality.

Our starting point is an exact comparison with a reference policy on the same tasks. Its surplus consists of quality

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change plus resource savings valued at the reference efficiency. The identity compares general router outcomes; its verified-cascade specialization exposes the costs of unsuccessful speculation and gives a homogeneous hit-rate threshold. We use it to derive an efficiency-aware residual-coverage criterion and a two-world lower bound for routers restricted to individual model scores. We then characterize the value of routing information and the consequences of correct writes, erroneous writes, and expiration in router memory. The memory model yields exact transient gains and distinct coverage laws for finite and heavy-tailed task spaces.

We characterize these router decisions through exact analysis and reproducible finite-model illustrations. Full proofs and extensions to latency, Markov hit dynamics, allocation curves, and statistical arm discovery appear in the appendix.

Relation to prior work. We build on the routing and cascading literature above, focusing on residual portfolio coverage and the records used to choose later routes. Narasimhan et al. (2025) connect cascades with speculative decoding and optimal deferral; our hit/miss accounting and organization around allocation, complementarity, and fallback are informed by SSD (Kumar et al., 2026). The mathematical tools are classical: fractional programming (Dinkelbach, 1967), information comparison (Blackwell, 1953), submodular coverage (Nemhauser et al., 1978), and infinite-urn occupancy (Karlin, 1967; Gnedin et al., 2007). We use them to connect routing costs, correlated capabilities, and persistent information within an explicit router model.

2. A Model of Inference Routers

Router policies and actions. An epoch is a decision with a measurable outcome. Its observable state x_n includes the task and available history; an information state I_n contains records from previous epochs. A router g selects an action and updates its information from feedback f_n:

$$u_n = g(x_n, I_n), \qquad I_{n+1} = \mathrm{Upd}(I_n, x_n, u_n, f_n). \quad (1)$$

The policy may randomize. An action specifies the selected models or tools, their roles, an optional aggregator, and an execution rule. A singleton router dispatches one capability; a portfolio router combines several. A verified cascade executes a primary allocation and uses its verification outcome to decide whether to invoke a fallback. An ex-ante router selects a path before execution. These rules incur different costs. For tasks whose intermediate states depend on the policy, we take the *entire task* as the epoch when comparing end-to-end routers; a comparison on internal states refers to that fixed state distribution.

Comparing routing policies. On a common task population, let (V, D) be the value and resource delivered and consumed under the router, and (Y_B, C_B) those of a reference policy. Resources include routing, execution, aggregation, verification, and information updates. Values and resources are nonnegative and integrable, with $\bar{c} = \mathbb{E}[C_B] > 0$ and $\mathbb{E}[D] > 0$. Write $\bar{v} = \mathbb{E}[Y_B]$ and $\eta_B = \bar{v}/\bar{c}$. For a jointly stationary ergodic sequence of these epoch variables, the long-run value per resource is

$$\eta = \lim_{N\to\infty} \frac{\sum_{n<N} V_n}{\sum_{n<N} D_n} = \frac{\mathbb{E}[V]}{\mathbb{E}[D]} \quad \text{almost surely.} \quad (2)$$

It is a ratio of aggregate quantities, not the average of V_n/D_n. The expectation comparison below also applies to a single epoch distribution without a stationarity claim.

Algorithm 1 A router with verified execution and escalation

1: Observe task state x and information state I.
2: Select a primary and fallback plan $u = g(x, I)$.
3: Execute the primary and verify its proposed output.
4: **if** the verification rule accepts the primary ($H = 1$) **then**
5: Commit the primary output.
6: **else**
7: Execute and commit the fallback.
8: **end if**
9: Record the outcomes and incurred resources; update I.

Theorem 1 (Reference-relative router surplus)**.** *Under the model above, define quality change Q and resource savings $\mathcal{S}$ by*

$$Q = \mathbb{E}[V - Y_B], \qquad \mathcal{S} = \mathbb{E}[C_B - D]. \quad (3)$$

Then $\mathbb{E}[V] = \bar{v} + Q$, $\mathbb{E}[D] = \bar{c} - \mathcal{S}$, and

$$\eta - \eta_B = \frac{Q + \eta_B \mathcal{S}}{\bar{c} - \mathcal{S}}. \quad (4)$$

Consequently, efficiency improves exactly when $Q + \eta_B \mathcal{S} > 0$. A quality constraint $\mathbb{E}[V] \geq \bar{v} - \epsilon$ is the separate condition $Q \geq -\epsilon$.

The identity follows by cross-multiplying the positive denominators. It gives a common comparison for singleton dispatch, portfolio composition, and cascade execution: quality change plus resource savings valued at the reference efficiency.

Verified execution with escalation. Algorithm 1 specializes the router to SCA. Let Y_P be primary value, $H \in \{0, 1\}$ the acceptance event, and (Y_F, C_F) the fallback value and *additional* resource if invoked. Let A include primary, routing, aggregation, verification, and update resources. Then

$$V = HY_P + (1 - H)Y_F, \qquad D = A + (1 - H)C_F. \quad (5)$$

The surplus terms become

$$\begin{aligned} Q &= \mathbb{E}[H(Y_P - Y_B) + (1-H)(Y_F - Y_B)]\,, \\ \mathcal{S} &= \mathbb{E}[HC_B - A - (1-H)(C_F - C_B)]\,. \end{aligned} \tag{6}$$

Here $\mathbb{E}[HC_B]$ is avoided reference work and A is the primary and routing bill. The remaining terms account for changes in delivered quality and recovery cost. No independence is required: accepting easy tasks can change reference costs and fallback quality conditional on rejection. The potential outcomes define the comparison; estimating them may require paired evaluation or additional calls. This specialization prices an unsuccessful primary attempt before fallback, whereas ex-ante dispatch pays only for its selected path and routing overhead.

The homogeneous cascade. For explicit decision rules, suppose the hit probability is p, the conditional hit and miss values are v_h, v_m, the primary bill is $a > 0$, and fallback costs $b > 0$. Then

$$\eta(p) = \frac{v_m + p(v_h - v_m)}{a + (1-p)b}. \tag{7}$$

With a matching fallback $v_m = \bar{v}$, $b = \bar{c}$, and $v_h, \bar{v} > 0$, SCA is at least as efficient as the reference exactly when

$$p \geq \frac{\bar{v}}{v_h}\frac{a}{b}. \tag{8}$$

Thus a tenfold primary price advantage requires at least 10% acceptance if accepted outputs preserve value. For $G = v_h/\bar{v}$ and $P = b/a$, the normalized efficiency is

$$\frac{\eta}{\eta_B} = \frac{pG + 1 - p}{1/P + 1 - p}. \tag{9}$$

The numerator is quality retention; the denominator is cost ratio. This separates a high-quality low-priced primary from an inexpensive but lossy one. For an ex-ante router that chooses one path before attempting any computation, the wasted primary term is absent. Its accounting is a different specialization, not (8).

3. What Should a Router Combine?

A portfolio router must determine which capability adds value to its current selection. Let E_m denote that candidate m supplies a sufficient output on a shared task. An exact selecting verifier succeeds whenever at least one selected candidate succeeds. For a portfolio S,

$$p(S) = \mathbb{P}\Big(\bigcup_{m \in S} E_m\Big), \qquad r_m(S) = \mathbb{P}\Big(E_m \cap \bigcap_{i \in S} E_i^c\Big). \tag{10}$$

The residual coverage $r_m(S)$ is the extra success probability supplied by m. It is defined under arbitrary correlations among candidate outputs. Conditional independence of model errors is unnecessary.

Theorem 2 (Efficiency-aware residual coverage)**.** *Suppose the compared portfolios have common conditional values v_h, v_m and fallback cost b, and use (7) with primary bill $a(S)$. Let $d_m(S) = a(S \cup \{m\}) - a(S)$, including any change in aggregation and verification cost. Adding m strictly improves efficiency if and only if*

$$r_m(S)\big[v_h - v_m + \eta(S)b\big] > \eta(S)d_m(S). \tag{11}$$

Adding the candidate increases delivered value by $r_m(S)(v_h - v_m)$ and changes cost by $d_m(S) - br_m(S)$. Substitution in the ratio gives (11). In the value-preserving case $v_h = v_m$, a candidate pays for itself exactly when its avoided fallback cost $br_m(S)$ exceeds its added bill. The marginal rule also prices a growing aggregator context. Maximizing coverage at a fixed budget is therefore one useful subproblem, rather than an unconditional replacement for maximizing efficiency.

Why a router needs joint outcomes. The identity $r_m(S) = p(S \cup \{m\}) - p(S)$ already shows that residual coverage is a joint property. The following lower bound makes the information gap explicit.

Proposition 3 (A limitation of marginal-quality selection)**.** *Consider a mandatory primary m_0 and a parallel choice of one additional candidate m_1 or m_2. All three candidates cost $c > 0$. A perfect fallback has cost $b > 0$ and all completed epochs have value one. There are two task distributions with identical individual success probabilities* $(1/2, 1/2, 1/2)$ *such that the optimal pair costs* $2c$*, but every possibly randomized selector whose input is only these marginal probabilities has worst-case expected cost at least* $2c + b/4$.

Let E_0 have probability $1/2$. In the first distribution, $E_1 = E_0$ and $E_2 = E_0^c$; in the second, exchange m_1, m_2. The complementary pair never escalates, whereas the redundant pair escalates on half of tasks. The marginal input is identical, so one of the two distributions makes the selector choose the redundant pair with probability at least $1/2$. Even exact individual scores cannot resolve this ambiguity. Paired outcomes or other information identifying residual coverage can.

Steering toward residual uncertainty. A role or context view can change where a candidate allocates its effort. To isolate that effect, let a task's accepted answer type $T^\star$ have law q on a finite set, and let two independent proposals have laws p_1, p_2, also independent of $T^\star$ conditional on the observable state. Fix a region T_0 to de-emphasize and write $T_1 = T_0^c$. For $\rho \in [0, 1]$, define

$$p_2^\rho(t) = \frac{p_2(t)}{Z(\rho)} \begin{cases} \rho, & t \in T_0, \\ 1, & t \in T_1, \end{cases} \qquad Z(\rho) = \rho W_0 + W_1, \tag{12}$$

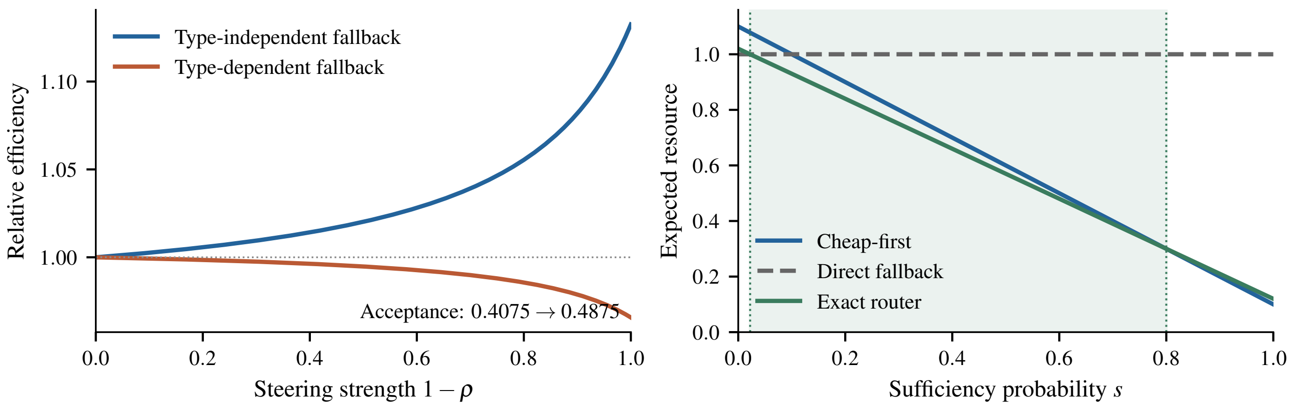

Figure 1. Exact finite-model calculations. **Left:** In the four-type construction, steering changes the pair's acceptance rate from 0.4075 to 0.4875 while keeping the second proposer's standalone quality at 0.25. Efficiency increases with a type-independent fallback but decreases with a fallback that succeeds only on types 3 and 4, despite the same rising acceptance rate. Each curve is normalized by its own unsteered efficiency. **Right:** For primary cost $a = 0.1$, reference cost $b = 1$, and exact router cost $k = 0.02$, buying routing information is optimal only for sufficiency probabilities between 0.0222 and 0.8. All values are calculated from the stated finite models.

where $W_k = \sum_{t \in T_k} p_2(t) > 0$. Lower ρ moves effort out of T_0. If $a_1 = \sum_t q(t)p_1(t) < 1$, the target law after primary failure is

$$q^-(t) = \frac{q(t)(1 - p_1(t))}{1 - a_1}, \qquad h(\rho) = \sum_t q^-(t) p_2^\rho(t). \tag{13}$$

Proposition 4 (Residual steering)**.** *For* $\bar{f}_k = W_k^{-1} \sum_{t \in T_k} f(t) p_2(t)$,

$$h'(\rho) = \frac{W_0 W_1}{Z(\rho)^2} \left(\bar{q}_0^- - \bar{q}_1^- \right). \tag{14}$$

The same identity with q in place of q^- gives the derivative of candidate 2's standalone quality. Hence reducing ρ strictly improves conditional coverage when $\bar{q}_1^- > \bar{q}_0^-$, even if it reduces standalone quality when $\bar{q}_0 > \bar{q}_1$.

This is a finite answer-type model, not an assumption about arbitrary language-model distributions. It isolates a distinction that marginal scoring misses. The pair success rate is $a_1 + (1 - a_1)h(\rho)$; applying Theorem 2 or differentiating (7) converts the coverage change into an efficiency comparison only when the compared values and costs satisfy their stated conditions. In particular, changing the failure set may change the fallback's conditional quality.

Coverage and allocation. Since the event newly covered by a candidate shrinks as the portfolio grows, $p(S)$ is monotone submodular for every fixed joint law of the E_m. The classical cardinality-constrained greedy guarantee therefore applies to maximizing this static coverage (Nemhauser et al., 1978). Prices require a knapsack treatment (Sviridenko, 2004). Observing failures during sequential selection can change conditional marginals, so a guarantee for an adaptive policy requires adaptive submodularity rather than static submodularity alone (Golovin & Krause, 2011). The distinction matters when an unsuccessful proposal identifies which complementary capability is needed.

4. When Should a Router Spend More?

Beyond choosing a portfolio, a router decides whether to attempt its selected primary, dispatch directly to a stronger path, or pay for information that changes this choice. Consider a state where the primary costs $a > 0$, is sufficient with probability p, and delivers value v_h on a hit. On a miss, fallback costs b on top of a and restores value v_b, the value of going directly to that path. For a value–cost objective $J_\lambda = \mathbb{E}[V - \lambda D]$, $\lambda > 0$, define

$$W_\lambda = \lambda b + v_h - v_b. \tag{15}$$

This quantity is the value of turning a miss into a hit: the avoided fallback bill, adjusted for the value difference.

Proposition 5 (Attempt-then-escalate threshold)**.** *Attempting the primary first is at least as good as direct fallback exactly when $pW_\lambda \geq \lambda a$. If $W_\lambda > 0$, this is $p \geq \lambda a / W_\lambda$; if $W_\lambda \leq 0$, attempting the primary is strictly worse. A routing refinement costing k that increases p by Δp while preserving the other quantities is worthwhile relative to the original attempt-then-escalate policy exactly when $\Delta p\, W_\lambda \geq \lambda k$.*

A threshold exceeding one simply means no available hit probability is sufficient. A quality premium $v_b - v_h$ raises the threshold, so price ratio alone is not a universal routing rule. At $\lambda = \eta_B$, an expected J_λ improvement is exactly a positive surplus in Theorem 1. More generally, the optimal ratio can be characterized through $\max_g \mathbb{E}[V_g - \lambda D_g]$ under the usual existence and positive-cost conditions of fractional programming (Dinkelbach, 1967). An external quality constraint must still be imposed separately.

Paying for a more informed routing decision. Suppose now that both accepted primary outputs and fallback preserve value. Let s be the current probability of primary sufficiency, $0 < a < b$, and suppose the router can acquire a signal revealing the sufficiency event exactly for cost $k \geq 0$. The router has three choices:

$$\begin{aligned} C_{\text{cascade}}(s) &= a + (1-s)b, \\ C_{\text{direct}}(s) &= b, \\ C_{\text{info}}(s) &= k + sa + (1-s)b. \end{aligned} \tag{16}$$

The informed choice avoids an unsuccessful primary attempt, while still paying for a successful one.

Proposition 6 (The routing-information band). *Exact routing information is strictly cheaper than both uninformed choices if and only if*

$$\frac{k}{b-a} < s < 1 - \frac{k}{a}. \tag{17}$$

This band is nonempty exactly when $k < a(b-a)/b$ and then contains the uninformed crossover $s = a/b$.

Near $s = 0$ there is little reason to test a likely failure; near $s = 1$ there is little reason to buy an additional routing signal. Extra routing information has the largest opportunity to change the action between these extremes. For an imperfect experiment Z with posterior s_Z, its net value is

$$\min\{C_{\text{cascade}}(s), b\} - \mathbb{E}\min\{C_{\text{cascade}}(s_Z), b\} - k. \tag{18}$$

The posterior has mean s, and exact revelation dominates any experiment about the same sufficiency event (Blackwell, 1953). Thus the perfect-information band bounds where such a router can pay for itself; it does not assert that a learned router achieves that bound. Figure 1 shows a numerical instance.

Parallelism changes the resource being optimized. Dollars or token charges generally add across calls. Concurrent latency is determined by completion times and dependencies. A primary and fallback launched simultaneously can reduce latency even while increasing expenditure, but cancellation may also save expenditure if only a small prefix of the fallback is billed. Both effects are represented by actual consumed resource, rather than by treating a launched call as necessarily free or necessarily fully billed. The appendix derives the corresponding latency criterion and a unique-crossing result for accurate versus fast fallback behind an independent barrier.

5. What Should a Router Remember?

A stateful router uses past outcomes to improve later choices through I in (1). Useful information may be a successful route, a task-conditioned capability estimate, or evidence about which candidates fail together. Verification supplies such records in our cascade setting. We model their decision value through the outcomes of informed and uninformed routing, allowing both erroneous writes and expiration. The central quantity is the resulting *quality–cost surplus*.

A record process. Partition tasks into J cells, arriving independently with probabilities $\nu_j > 0$. At the start of an epoch, each cell is uninformed (U), holds an informative record (I), or holds an erroneous record (W). After the epoch's decision, each existing record expires independently with probability δ_j. The arriving cell then receives an informative write with probability ω_j or an erroneous write with probability ψ_j; a new write replaces its old record. Write types are independent of the past and of the current record. Set

$$r_j = \omega_j + \psi_j > 0, \qquad \alpha_j = (1-\delta_j)(1-\nu_j r_j) < 1. \tag{19}$$

The timing matters: a record written after epoch n first affects epoch $n+1$. Let v_j^s, c_j^s be the router's conditional expected value and cost in cell j, record state $s \in \{U, I, W\}$, including the computation used to update the record. These fixed outcomes summarize how the router uses each record state.

Measuring a record by surplus. The cold reference router, which collects records at the same cost but always makes an uninformed allocation, has efficiency

$$\eta_U = \frac{\sum_j \nu_j v_j^U}{\sum_j \nu_j c_j^U}. \tag{20}$$

Assume its denominator and the stationary expected cost below are positive. Define the reference-relative effect of each record by

$$g_j^s = (v_j^s - v_j^U) - \eta_U(c_j^s - c_j^U), \quad s \in \{I, W\}. \tag{21}$$

An informative record can save cost as well as improve value; an erroneous one can incur either kind of loss. Let

$$G_j = \frac{\nu_j(\omega_j g_j^I + \psi_j g_j^W)}{1 - \alpha_j}. \tag{22}$$

Theorem 7 (Transient and stationary memory gains). *From an empty store, the probabilities of informative and erroneous records before epoch $n \geq 0$ are*

$$i_{j,n} = \frac{\nu_j \omega_j}{1-\alpha_j}(1-\alpha_j^n), \qquad w_{j,n} = \frac{\nu_j \psi_j}{1-\alpha_j}(1-\alpha_j^n). \tag{23}$$

The stationary efficiency satisfies $\eta_\infty \geq \eta_U$ if and only if $\sum_j \nu_j G_j \geq 0$. At every horizon N,

$$\sum_{n=0}^{N-1} \mathbb{E}[V_n - \eta_U D_n] = \sum_j \nu_j G_j \left[N - \frac{1-\alpha_j^N}{1-\alpha_j} \right]. \tag{24}$$

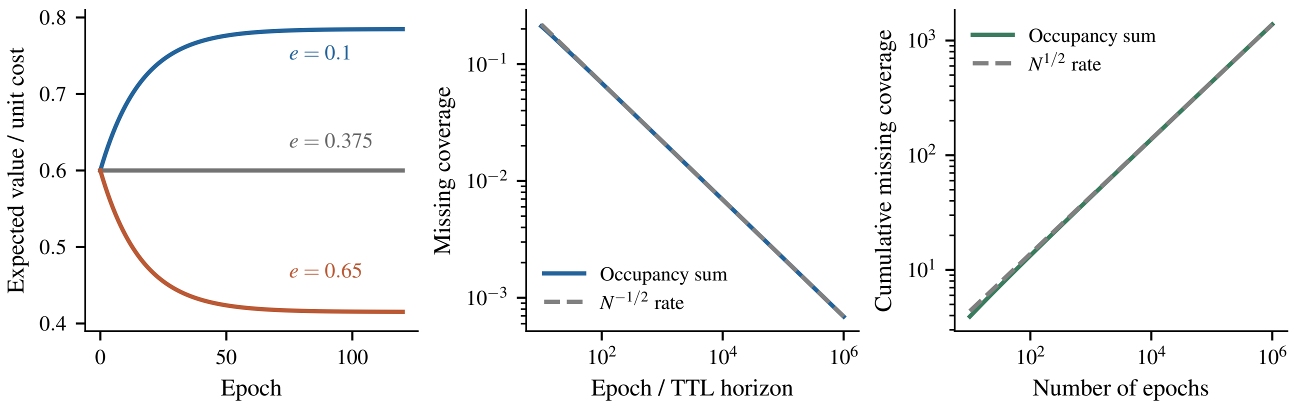

Figure 2. Exact consequences of the record model. **Left:** Ten equally likely cells, total per-visit write rate 0.5, per-epoch expiration probability 0.01, and unit cost in every record state. Values are $(v^U, v^I, v^W) = (0.6, 0.9, 0.1)$. Curves are labeled by erroneous fraction e; informative writes improve the cold policy only when $e < 0.375$; faster convergence alone would not change this condition. **Center/right:** For clean records, Zipf exponent $z = 2$, and informed-minus-cold hit gap one, the missing coverage scales as $N^{-1/2}$ while cumulative missing coverage scales as $N^{1/2}$. Solid curves are occupancy calculations; dashed curves show their asymptotic rates.

If $g_j^I > 0 > g_j^W$, cell j contributes positive stationary surplus exactly when

$$\frac{\psi_j}{r_j} < \frac{g_j^I}{g_j^I - g_j^W}. \tag{25}$$

To see the recurrence, a good record is either freshly written, with probability $\nu_j\omega_j$, or survives both expiration and the absence of a new write. Thus $i_{j,n+1} = \nu_j\omega_j + \alpha_j i_{j,n}$, and similarly for $w_{j,n}$. Solving these recurrences and summing the surplus yields the theorem. The marginal convergence rate is at most $\max_j \alpha_j < 1$ for finitely many cells. The persistent store can correlate successive hits; this derivation does not identify the hit sequence with a two-state Markov chain.

The threshold (25) explains why more records need not improve decisions. Write quality determines the sign of each cell's contribution; write frequency and expiration determine its stationary weight and convergence rate. With heterogeneous cells, these weights can also change the sign of the total gain. The transient weights in (24) also differ, so aggregate improvement need not be monotone in time even when the final gain is positive. The formula measures the ratio of expected aggregate outcomes through its surplus; it is not a claim about the expectation of a finite-sample random ratio.

Against a reference that does not maintain a store, the maintenance bill must also be recovered. For any reference efficiency λ, let $J_U(\lambda) = \sum_j \nu_j(v_j^U - \lambda c_j^U)$ and evaluate the record gains in Equation (21) at λ. The stationary condition becomes

$$J_U(\lambda) + \sum_j \nu_j G_j(\lambda) \geq 0. \tag{26}$$

If the cold-state process has the same value as the no-memory reference but pays an extra maintenance cost h per epoch, then $J_U(\lambda) = -\lambda h$. Thus the records must deliver at least λh in additional surplus. Appendix G.2 gives the finite-horizon form.

5.1. Capacity and the Tail of Task Frequencies

Memory capacity does not in general behave like deterministic expiration. If a record expires after E global epochs, stationary coverage of cell j is

$$\chi_j^{\mathrm{TTL}} = 1 - (1 - \nu_j r_j)^E. \tag{27}$$

For an ordinary FIFO containing the last M write events, including duplicate cell records, let $\beta = \sum_j \nu_j r_j$ and consult the newest matching record. Its stationary coverage is instead

$$\chi_j^{\mathrm{FIFO}} = 1 - \left(1 - \frac{\nu_j r_j}{\beta}\right)^M. \tag{28}$$

Both have informative fraction ω_j/r_j conditional on coverage. The FIFO formula follows by conditioning on the write stream. Replacing the random span of M writes by its mean M/β inside (27) is not an exact calculation.

For clean writes, the uncovered hit deficit at TTL horizon E is $\Delta \sum_j \nu_j(1 - \omega\nu_j)^E$.

Theorem 8 (Heavy-tailed information deficit)**.** *Suppose cells are countably infinite with $\nu_j = j^{-z}/\zeta(z)$, $z > 1$. Writes are clean with constant probability $\omega \in (0, 1]$, and every cell has the same positive informed-minus-cold hit gain Δ. Under a TTL of E epochs, the stationary uncovered hit deficit is $\Theta(E^{-(z-1)/z})$. From an empty store without expiration, the cumulative uncovered hit deficit over N epochs is $\Theta(N^{1/z})$.*

In an infinite task space, increasingly rare cells keep arriving. There is no uniform geometric rate, and the cumulative loss from missing information need not flatten to a constant. These occupancy exponents (Karlin, 1967; Gnedin et al., 2007) describe repeated access to transferable records. Their allocation interpretation is nevertheless useful: a task family with a heavier tail may continue to benefit from capacity after a concentrated family has saturated.

5.2. Allocating Router Memory Across Task Families

Router memory raises a resource-allocation question across task families. Suppose K independent positions or families have increasing, strictly concave differentiable coverage curves $p_k(F_k)$, where $F_k \geq 0$ is a continuous allocation and $\gamma_k > 0$ its unit price. Let $w_k > 0$ be the value of coverage at position k. For J_λ, this weight can include the family frequency multiplied by the margin W_λ in (15); nonpositive-margin speculation is excluded. A hard-budget subproblem is

$$\max_{F \geq 0} \sum_{k=1}^{K} w_k p_k(F_k) \quad \text{subject to} \quad \sum_k \gamma_k F_k \leq B. \tag{29}$$

This is separable allocation across positions, not a model of mutually redundant proposers within one position.

Proposition 9 (Coverage allocation)**.** *For $B > 0$, (29) has a unique optimizer. There is a multiplier $\Lambda > 0$ such that funded positions satisfy $w_k p_k'(F_k) = \Lambda \gamma_k$, and unfunded positions satisfy $w_k p_k'(0) \leq \Lambda \gamma_k$. If $1 - p_k(F) = d_k(1 + F)^{-r}$ with $d_k \in (0, 1]$ and common $r > 0$, then*

$$F_k^\star = \left[\mu \left(\frac{w_k d_k}{\gamma_k} \right)^{1/(1+r)} - 1 \right]_+, \tag{30}$$

where μ makes the budget bind.

This is the standard marginal-equalization principle applied to the value of coverage. Theorem 8 motivates an asymptotic exponent $r = 1 - 1/z$ for clean Zipf families; it does not make the shifted power law exact at finite capacity. With common z and large funded allocations, (30) assigns capacity in proportion to $(w_k d_k / \gamma_k)^{z/(2z-1)}$. More valuable residual uncertainty and lower unit price attract capacity, while saturation limits concentration. Exponential coverage curves lead instead to logarithmic water filling, derived in the appendix.

6. Empirical Checks on a Real Outcome Matrix

Data and protocol. CodeRouterBench records an execution-verified score in $[0, 1]$, billed dollars, and latency for 9,999 coding tasks under each of 8 models, so (Y_B, C_B) and (Y_P, A) of Section 2 are observed for every task. The reference B is always `claude-opus-4-6`. The cascades below instantiate the primary as one lower-priced model (Algorithm 1 allows any allocation; the data support this instance), accept with the exact verifier $H = \mathbf{1}\{\text{score} \geq \theta\}$, $\theta \in \{1, 0.5\}$, and commit Opus on a miss, so $(Y_F, C_F) = (Y_B, C_B)$. Primitives come from the 7,080-task probing split, outcomes and 95% bootstrap CIs from the disjoint 2,919-task `id_test` split. The benchmark has no arrival order, so streams, records, hit chains, barrier rounds, and Zipf frequencies are constructions over real outcomes. Appendix J lists every claim, number, and script.

Surplus and the acceptance floor. Theorem 1 and (32) hold to residual 10^{-14}. The matching hypothesis of (8) fails: acceptance selects tasks Opus also solves ($\bar{v}_h / \bar{v} \in [1.58, 2.08]$), so the CI of $v_m / \bar{v}$ excludes 1 in 28/28 configurations and that of $b / \bar{c}$ in 24/28. The stated floor gives the sign of $\eta - \eta_B$ in 19/28 pooled configurations, and every miss is a predicted win that loses. Out of sample (9) over-predicts in 13/14 cascades (median $+15.3\%$), while the state-matched form of Appendix A.3 errs by 2.7% and covers the realization in 12/14 (Figure 3a). The ex-ante formula of Appendix A.2 under-predicts nine released routers by 32–49%.

Residual coverage. The common-value hypothesis of Theorem 2 fails in the optimistic direction: v_m falls in 97% of 1,457 non-trivial (S, m) pairs. The sign of (11) with S's values still matches the realized sign in 90.3%/93.3% of pairs at $\theta = 1/0.5$ and in 91.4%/93.1% out of sample, against 65–72% for the marginal-only proxy $(1 - p(S))\mathbb{P}(E_m)$; the top partner by (11) beats the top-$\mathbb{P}(E_m)$ partner in 58/58 and 114/114 portfolios. Errors are positively correlated in 99% of model pairs, so the gap of Proposition 3 appears in only 5–11% of contexts (held-out $+0.032$ $[0.020, 0.042]$) and the adaptive counterexample of Appendix E.6 is NOT INSTANTIABLE.

Thresholds and the value of information. Proposition 5 as a sign rule agrees with the realized sign of $J_{\text{casc}} - J_{\text{direct}}$ on 92.5%/93.3% of decided 9-/21-state cells (Figure 3b), but $\lambda a / W_\lambda \leq 0.05$ in every pooled state, so $p \geq a/b$ scores 0.95/0.92. Its magnitude fails: pooled Qwen3-Max ($\theta = 1$, $\lambda = 1$) predicts 0.124 $[0.120, 0.129]$ and realizes 0.071 $[0.062, 0.080]$, because $\mathbb{E}[Y_B \mid H = 0] - \mathbb{E}[Y_B] = -0.073$ $[-0.083, -0.064]$. The band (17) picks the realized cheapest option in 96.7%/99.0% of decided cells. No router's saving exceeds $f(s) - g(s)$ (largest ratio 0.952); at $\theta = 1$ with Qwen3-Max the fine-tuned routers capture 0.45–0.59 of it and the classical routers -1.1 to -1.9.

Records, retention, and heavy tails. With an $\varepsilon = 0.25$ exploration logger writing verified records, (23) and (24)

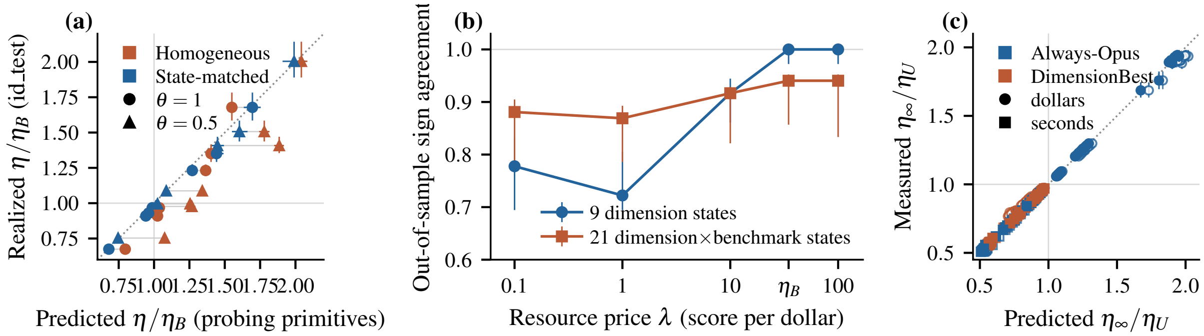

Figure 3. Checks on CodeRouterBench (probing-split primitives, `id_test` outcomes, 95% bootstrap CIs). **(a)** Realized η/η_B of 14 verified cascades ($\theta = 1$ circles, $\theta = 0.5$ triangles, Opus fallback) against the prediction of the homogeneous form (9) under $v_m = \bar{v}$, $b = \bar{c}$ (red) and of the state-matched form of Appendix A.3 (blue); dotted line $y = x$. **(b)** Proposition 5 as an out-of-sample sign rule: fraction of 9 dimension states (blue) and 21 dimension×benchmark states (red) in which $\mathrm{sign}(pW_\lambda - \lambda a)$ from probing equals the realized sign of $J_{\text{casc}} - J_{\text{direct}}$, $\lambda \in \{0.1, 1, 10, \eta_B, 100\}$. **(c)** Stationary criterion of Theorem 7: measured η_∞/η_U against $1 + (\sum_j \nu_j G_j / c_\infty)/\eta_U$ for 128 record-process configurations (filled: in-population primitives; open: probing primitives).

match the stream (98.1% of bins within 2 SE; paired CI covers 96.7% of points), and the stationary criterion of Theorem 7 predicts the sign of $\eta_\infty - \eta_U$ in 128/128 configurations (Figure 3c), which form four sign cases: against always-Opus, records raise dollar efficiency from 42.9 to 72–83 score per dollar but lower it in seconds; against DimensionBest $g^I_j < 0$ in most cells and memory hurts. In the 3–5 of 21 cells that meet $g^I_j > 0 > g^W_j$, (25) is right in 84/84 and 76/79 decisions. (46) and (47) are covered in 96.8%/94.8% of points and $E \to M/\beta$ is rejected 12/12. Constructed Zipf frequencies give the exponents of Theorem 8 as -0.504 and 0.503 at $z = 2$ (predicted -0.5 and 0.5).

Hit dynamics, hedging, barrier, and allocation. Proposition 10 behaves as stated on constructed Markov streams (transient $\max|z| \leq 2.2$), and a hidden three-regime stream keeps flow balance but rejects the transient ($\max|z|$ 15–44), as Appendix B warns. Appendix D.2 does not apply here: Opus is the fastest model, so $\ell_p \leq \ell_b$ fails in 14/14 cascades. For Proposition 13 existence agrees in 6/8 pairs (2 inconclusive), but b^* misses the empirical CI in 3/3 pairs because round latency is not deterministic. The hypotheses of Proposition 11 fail on price-ordered curves (marginal gains rise at 3/9 positions), the closed form (36) loses 11.4% against the exact optimum, while marginal equalization with per-position fits is within -0.0104 $[-0.0121, -0.0087]$ of it and $+0.0302$ $[+0.0272, +0.0336]$ above uniform.

What these checks do not establish. Identities can fail only through implementation; the findings are the measured hypothesis violations and the out-of-sample checks. Every FAIL rejects a stated hypothesis on this data; none is a counterexample to a theorem. The hit label is the benchmark's own score column, so at $\theta = 1$ a hit has $Y_P = 1 \geq Y_B$ by construction, and a real verifier would add false accepts and rejects. Proposition **??** is untested.

7. Design Principles for Inference Routers

The theory connects three decisions made by an inference router. It should combine capabilities according to the failures they resolve jointly, spend on a stronger routing signal when the expected decision improvement pays for that signal, and retain records according to their effect on future routing surplus. A single model score cannot determine all three. The reference-relative comparison gives these decisions a common quality–cost objective, with explicit criteria for verified execution with escalation.

The results also identify what has to be measured. A matched reference and consumed resources determine the surplus in Theorem 1; paired candidate outcomes identify (10); write frequencies and record-conditioned outcomes determine Theorem 7. A verifier score alone supplies none of these comparisons automatically. In the record model, an informative write is a stipulated transfer of useful information. Learning its value from noisy finite observations requires exploration and estimation; the appendix distinguishes discovering an arm from estimating its mean. Extending the model to policy-dependent writing, learned task partitions, and nonstationary capability pools is a statistical control problem beyond these fixed-process results.

Inference routing is a problem of selecting computation and managing the information used to select it. Singleton dispatch, portfolio composition, and verified cascades expose different parts of that problem. The results show how complementarity, decision cost, and memory determine whether a router improves on its reference, and identify the joint outcomes and record values needed to make those decisions.

Impact Statement

This paper studies inference routers that select and combine models and tools, pay for routing information, and retain feedback for later decisions. Lower inference cost can make capable systems more accessible, while repeated verification and additional proposals can increase total resource consumption. The analysis separates resource savings from output quality and identifies how erroneous records can degrade later routing decisions. Applying the results requires measuring the relevant task outcomes and costs; an improvement in value per resource alone does not establish that an application meets its required quality level.

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A. Router Comparison and Homogeneous Cascades

A.1. Proof of Theorem 1

Proof. For any router, the definitions $Q = \mathbb{E}[V - Y_B]$ and $\mathcal{S} = \mathbb{E}[C_B - D]$ give $\mathbb{E}[V] = \bar{v} + Q$ and $\mathbb{E}[D] = \bar{c} - \mathcal{S}$. Under verified execution with escalation, Equation (5) further gives the pointwise expansions

$$V - Y_B = H(Y_P - Y_B) + (1 - H)(Y_F - Y_B),$$

and

$$C_B - D = HC_B - A - (1 - H)(C_F - C_B).$$

Taking expectations yields Equation (6). In either case, $\bar{c} - \mathcal{S} > 0$ and

$$\eta - \eta_B = \frac{\bar{v} + Q}{\bar{c} - \mathcal{S}} - \frac{\bar{v}}{\bar{c}} = \frac{Q + (\bar{v}/\bar{c})\mathcal{S}}{\bar{c} - \mathcal{S}}.$$

The denominator is positive, giving the efficiency comparison. The quality condition follows separately from $\mathbb{E}[V] = \bar{v} + Q$. These equalities require neither independence nor stationarity. For the long-run statement, the ergodic theorem applied to integrable V_n and D_n gives convergence of their sample means to their expectations. Their ratio converges because the limiting mean cost is positive. □

The comparison uses a reference policy on the same task population. If routing changes subsequent within-task states, taking the entire task as the epoch includes those consequences in (V, D); under cascade execution this includes the continuation represented by (Y_F, C_F). Evaluation on intermediate states visited by the router instead gives a comparison on that particular state distribution.

For example, if fallback restores the reference on every miss, values lie in $[0, 1]$, and $e = \mathbb{P}(H = 1, Y_P < Y_B)$, then $Q \geq -e$. Savings satisfying $\eta_B \mathcal{S} \geq e$ suffice for an efficiency guarantee, while $e \leq \epsilon$ suffices for the quality constraint. Neither conclusion requires the verifier to preserve the reference distribution.

A.2. Homogeneous threshold and endpoint bounds

Total expectation gives

$$\eta(p) = \frac{pv_h + (1 - p)v_m}{a + (1 - p)b}.$$

For matched fallback $v_m = \bar{v}$, $b = \bar{c}$, with $v_h, \bar{v}, a, b > 0$,

$$\begin{aligned} \eta(p) \geq \eta_B &\iff b[pv_h + (1 - p)\bar{v}] \geq \bar{v}[a + (1 - p)b] \\ &\iff pbv_h \geq \bar{v}a. \end{aligned}$$

Strict comparison follows by replacing both inequalities with strict inequalities. A floor above one cannot be attained by an admissible acceptance probability.

For $G = v_h/\bar{v}$ and $P = b/a$, write

$$R(p) = \frac{pG + 1 - p}{1/P + 1 - p}, \qquad R'(p) = \frac{G(1 + P) - 1}{P(1/P + 1 - p)^2}.$$

Since the derivative has constant sign and the endpoint values are $R(0) = P/(1 + P)$ and $R(1) = GP$,

$$\min\left\{GP, \frac{P}{1 + P}\right\} \leq R(p) \leq \max\left\{GP, \frac{P}{1 + P}\right\}, \quad 0 \leq p \leq 1. \tag{31}$$

Also $R(p) \geq [P/(1 + P)](pG + 1 - p)$ because its denominator is at most $(1 + P)/P$. When $G(1 + P) < 1$, the ratio decreases and remains below one. The crossing $R(p) = 1$ is $p = 1/(GP)$, feasible exactly when $GP \geq 1$.

More generally, holding conditional values and positive resources fixed,

$$\frac{d}{dp} \frac{pv_h + (1 - p)v_m}{pd_h + (1 - p)d_m} = \frac{v_h d_m - v_m d_h}{[pd_h + (1 - p)d_m]^2}.$$

Thus higher acceptance improves efficiency precisely when a hit has greater value per resource than a miss. Changing the accepted task population need not preserve these conditional means.

For an ex-ante router, let T indicate choosing the primary branch before execution and $t = \mathbb{P}(T)$. With no routing overhead, primary cost a, reference cost b, and conditional values $v_h, \bar{v}$, normalized efficiency is

$$\frac{tG + 1 - t}{ta/b + 1 - t}.$$

The always-paid wasted-primary term is absent. Branch selection T is not the verified acceptance event H.

A.3. State-matched reference values and costs

Suppose fallback restores the reference on misses: $Y_F = Y_B$ and $C_F = C_B$ there. Let $\bar{v}_h, \bar{v}_m$ and $\bar{c}_h, \bar{c}_m$ denote the reference value and cost means conditional on $H = 1, 0$, respectively. Let $v_h = \mathbb{E}[Y_P \mid H = 1]$ and $a = \mathbb{E}[A]$. For $0 < p < 1$,

$$Q = p(v_h - \bar{v}_h), \qquad \mathcal{S} = p\bar{c}_h - a,$$

while $\bar{v} = p\bar{v}_h + (1-p)\bar{v}_m$ and $\bar{c} = p\bar{c}_h + (1-p)\bar{c}_m$. The exact efficiency condition is

$$p(v_h - \bar{v}_h) + \eta_B(p\bar{c}_h - a) \geq 0. \tag{32}$$

In particular, $v_h \geq \bar{v}_h$ and $a \leq p\bar{c}_h$ suffice. If $\eta_B > 0$, either strict inequality yields strict improvement. The event H may be random conditional on the observable state.

The same matched fallback gives

$$\frac{\mathbb{E}[HC_B]}{\mathbb{E}[C_B]} = 1 - \frac{\mathbb{E}[D]}{\mathbb{E}[C_B]} + \frac{\mathbb{E}[A]}{\mathbb{E}[C_B]}.$$

Therefore cost savings lower-bound a fraction weighted by reference cost. They lower-bound ordinary acceptance frequency only under an additional condition such as constant C_B.

A heavy router invoked before commitment need not have an invocation event independent of H. Actual overhead is included through $a = \mathbb{E}[A]$. If primary plus gate cost $c_p + \kappa_g$ is paid every epoch and heavy-router cost κ_h only on misses, homogeneous matched fallback instead gives

$$\eta \geq \eta_B \iff p \geq \frac{\bar{v}(c_p + \kappa_g + \kappa_h)}{bv_h + \bar{v}\kappa_h}.$$

This follows by the same cross-multiplication and incorporates the dependence of average overhead on the miss rate.

B. An Optional Two-State Model of Hit Dynamics

Proposition 10 (Two-state hit dynamics)**.** *Suppose $(H_n)_{n\geq 1}$ is a homogeneous Markov chain with transition probabilities $p_h = \mathbb{P}(H_{n+1} = 1 \mid H_n = 1)$ and $p_m = \mathbb{P}(H_{n+1} = 1 \mid H_n = 0)$, where $0 < p_m \leq 1$, $0 \leq p_h < 1$, and $(p_h, p_m) \neq (0, 1)$. Then, with $\xi = p_h - p_m$,*

$$\pi = \frac{p_m}{1 - p_h + p_m}, \qquad \mathbb{P}(H_n = 1) = \pi + (\mathbb{P}(H_1 = 1) - \pi)\xi^{n-1}.$$

Let $\mathcal{F}_n$ include all hit labels, values, and resources through epoch n. Suppose

$$\mathbb{E}[V_n \mid \mathcal{F}_{n-1} \vee \sigma(H_n)] = v_{H_n}, \qquad \mathbb{E}[D_n \mid \mathcal{F}_{n-1} \vee \sigma(H_n)] = d_{H_n},$$

with $d_0, d_1 > 0$ and $\sup_n \mathbb{E}[V_n^2 + D_n^2] < \infty$. Then almost surely

$$\lim_{N\to\infty} \frac{\sum_{n=1}^N V_n}{\sum_{n=1}^N D_n} = \frac{\pi v_1 + (1-\pi)v_0}{\pi d_1 + (1-\pi)d_0}.$$

Proof. Writing $\pi_n = \mathbb{P}(H_n = 1)$ gives $\pi_{n+1} = p_m + (p_h - p_m)\pi_n$. Solving its fixed-point recurrence proves the first formula, with $|\xi| < 1$ under the stated restrictions. The finite irreducible chain has almost-sure empirical state frequencies $(1 - \pi, \pi)$.

The centered increments $Z_n = V_n - v_{H_n}$ satisfy $\mathbb{E}[Z_n \mid \mathcal{F}_{n-1}] = 0$ by the tower property. The uniform second-moment bound gives $\sum_n \mathbb{E}[Z_n^2]/n^2 < \infty$. The martingale $\sum_{n \le N} Z_n/n$ therefore converges almost surely, and Kronecker's lemma yields $N^{-1} \sum_{n \le N} Z_n \to 0$. Repeat for $D_n - d_{H_n}$ and combine with the state frequencies. The limiting denominator is positive. □

A stationary binary process already satisfies the flow balance $(1 - \pi)p_m = \pi(1 - p_h)$. The stationary expression alone does not demonstrate the Markov property. The transient formula is an extra model prediction; in particular, the record process in the main text does not generally have a Markov binary hit projection.

C. Thresholds and the Value of Routing Information

C.1. Proof of Proposition 5

Proof. The direct path has objective $J_{\text{direct}} = v_b - \lambda b$. Attempting the primary first has

$$J_{\text{cascade}} = pv_h + (1 - p)v_b - \lambda[a + (1 - p)b].$$

Subtracting,

$$J_{\text{cascade}} - J_{\text{direct}} = p(v_h - v_b + \lambda b) - \lambda a = pW_\lambda - \lambda a.$$

This proves the comparison and, for $W_\lambda > 0$, the threshold. If $W_\lambda \le 0$, then the difference is strictly negative because $\lambda, a > 0$.

A routing refinement whose stipulated effects are to change p to $p + \Delta p$ and add cost k changes the attempt-then-escalate objective by exactly

$$\Delta p(v_h - v_b + \lambda b) - \lambda k = \Delta p\, W_\lambda - \lambda k.$$

Its sign proves the second assertion. This comparison changes the success probability; it does not model a signal allowing contingent choice between fixed actions. □

The fractional-programming connection follows from

$$\mathbb{E}[V_g - \lambda D_g] = \mathbb{E}[D_g](\eta(g) - \lambda).$$

If an optimal ratio η^* is attained and expected costs are uniformly bounded away from zero, the supremum of the left side over g is zero at $\lambda = \eta^*$, positive below, and negative above, under the usual finiteness conditions. Finite exogenous states and finite action sets with positive expected costs satisfy these conditions; maximization then separates state by state. This is the standard parametric characterization of fractional programming (Dinkelbach, 1967).

C.2. Proof of Proposition 6

Proof. Subtract informed cost from each uninformed alternative:

$$C_{\text{direct}}(s) - C_{\text{info}}(s) = s(b - a) - k, \qquad C_{\text{cascade}}(s) - C_{\text{info}}(s) = a(1 - s) - k.$$

Both differences are strictly positive exactly when $k/(b - a) < s < 1 - k/a$. The interval is nonempty precisely when

$$\frac{k}{b - a} < 1 - \frac{k}{a} \iff k < \frac{a(b - a)}{b}.$$

This same condition places a/b strictly between the two endpoints. Since $k \ge 0$, the band lies within $0 < s < 1$. Equality at an endpoint means that information ties one uninformed alternative. □

C.3. Imperfect information and the perfect-information bound

Let U denote the binary sufficiency event, with $\mathbb{P}(U = 1) = s$. An experiment Z produces posterior $s_Z = \mathbb{P}(U = 1 \mid Z)$. Put

$$f(s) = \min\{a + (1-s)b, b\}, \qquad g(s) = sa + (1-s)b.$$

After paying for Z, choosing the cheaper available action has expected cost $k + \mathbb{E}[f(s_Z)]$. Thus its net saving is exactly $f(s) - \mathbb{E}[f(s_Z)] - k$.

The minimum of two affine functions is concave. Since $\mathbb{E}[s_Z] = s$, Jensen's inequality gives $\mathbb{E}[f(s_Z)] \le f(s)$. Also $f(t) \ge g(t)$ for every $t \in [0, 1]$: cascade minus $g(t)$ is $a(1-t)$, and direct minus $g(t)$ is $t(b-a)$. Since g is affine,

$$g(s) = \mathbb{E}[g(s_Z)] \le \mathbb{E}[f(s_Z)] \le f(s).$$

Gross information value is therefore at most $f(s) - g(s)$. Exact revelation attains this bound because $f(0) = g(0) = b$ and $f(1) = g(1) = a$. This proves directly the relevant perfect-information comparison (Blackwell, 1953). In particular,

$$f(s) - g(s) = \min\{s(b-a), a(1-s)\},$$

maximized at $s = a/b$ with value $a(b-a)/b$. At the same experiment cost k, no imperfect experiment strictly reduces cost outside the perfect-information band. Experiments with different costs require separate comparisons.

D. Verification and Parallel Resource Accounting

D.1. What a calibrated acceptance event guarantees

Let Y indicate primary correctness and let $\mathcal{G}$ contain all information used for commitment, including controller randomization. Suppose $v = \mathbb{E}[Y \mid \mathcal{G}]$ and the measurable event $\{H = 1\}$ is contained in $\{v \ge \theta\}$. For a positive-probability acceptance event,

$$\mathbb{P}(Y = 1 \mid H = 1) = \mathbb{E}[v \mid H = 1] \ge \theta.$$

For a nonnegative integrable $\mathcal{G}$-measurable stake w,

$$\mathbb{E}[wY \mid H = 1] = \mathbb{E}[wv \mid H = 1] \ge \theta \mathbb{E}[w \mid H = 1].$$

These conclusions require calibration relative to all selection information. A score calibrated only marginally need not retain calibration after selection using additional predictive information. The surplus theorem does not assume calibration.

D.2. Hedged latency and cancellation charges

Launch primary and reference concurrently after controller overhead ℓ_κ. Suppose deterministic completion times satisfy $0 < \ell_p \le \ell_b$, with primary verification included in ℓ_p, and the controller waits for the primary verdict. It terminates at $\ell_\kappa + \ell_p$ on a hit and $\ell_\kappa + \ell_b$ on a miss. With hit value v_h and matched miss/reference value $\bar{v} > 0$,

$$\eta^{\ell}_{\text{hedge}} = \frac{pv_h + (1-p)\bar{v}}{\ell_\kappa + p\ell_p + (1-p)\ell_b},$$
$$\eta^{\ell}_{\text{hedge}} \ge \frac{\bar{v}}{\ell_b} \iff p(\ell_b v_h - \bar{v}\ell_p) \ge \bar{v}\ell_\kappa. \tag{33}$$

This follows by multiplying the two positive denominators and canceling the common miss term. If $\ell_\kappa = 0$, $v_h \ge \bar{v}$, and $\ell_p \le \ell_b$, the comparison holds, strictly when $p > 0$ and either component comparison is strict. Sequential escalation instead has mean latency $\ell_\kappa + \ell_p + (1-p)\ell_b$ and condition $p\ell_b v_h \ge \bar{v}(\ell_\kappa + \ell_p)$.

For dollars, let $a > 0$ be the primary/controller bill, $b > 0$ the full reference bill, and r the conditional expected charge for a prefix consumed before cancellation on a hit. Then

$$\eta^{c}_{\text{hedge}} = \frac{pv_h + (1-p)\bar{v}}{a + (1-p)b + pr}, \qquad \eta^{c}_{\text{hedge}} \ge \frac{\bar{v}}{b} \iff p(bv_h - \bar{v}r) \ge \bar{v}a. \tag{34}$$

Launching the fallback does not imply full billing. For example, $\bar{v} = v_h = b = 1$, $a = r = 0.1$, and $p = 0.9$ give expected cost 0.29 and value one. If $r = b$, mean cost is $a + b$, exceeding reference cost; sufficiently higher hit value can nevertheless improve efficiency.

For random completion times, compute mean latency from realized times and acceptance. In general $\mathbb{E}[\max(L_p, L_b)] \neq \max(\mathbb{E}[L_p], \mathbb{E}[L_b])$. The corresponding conditional resource means can replace the deterministic times without changing Theorem 1.

E. Coverage Allocation Across Independent Positions

E.1. Setting and statement

Suppose K independent positions or task families have increasing, strictly concave differentiable coverage curves $p_k(F_k)$, where $F_k \geq 0$ is a continuous allocation and $\gamma_k > 0$ its unit price. Let $w_k > 0$ be the value of coverage at position k. For J_λ, this weight can include the family frequency multiplied by the margin W_λ in (15); nonpositive-margin speculation is excluded. A hard-budget subproblem is

$$\max_{F \geq 0} \sum_{k=1}^{K} w_k p_k(F_k) \quad \text{subject to} \quad \sum_k \gamma_k F_k \leq B. \tag{35}$$

This is separable allocation across positions, not a model of mutually redundant proposers within one position.

Proposition 11 (Coverage allocation)**.** *For $B > 0$, (35) has a unique optimizer. There is a multiplier $\Lambda > 0$ such that funded positions satisfy $w_k p_k'(F_k) = \Lambda \gamma_k$, and unfunded positions satisfy $w_k p_k'(0) \leq \Lambda \gamma_k$. If $1 - p_k(F) = d_k(1 + F)^{-r}$ with $d_k \in (0, 1]$ and common $r > 0$, then*

$$F_k^\star = \left[\mu \left(\frac{w_k d_k}{\gamma_k} \right)^{1/(1+r)} - 1 \right]_+, \tag{36}$$

where μ makes the budget bind.

This is the standard marginal-equalization principle applied to the value of coverage. Theorem 8 motivates an asymptotic exponent $r = 1 - 1/z$ for clean Zipf families; it does not make the shifted power law exact at finite capacity. With common z and large funded allocations, (36) assigns capacity in proportion to $(w_k d_k / \gamma_k)^{z/(2z-1)}$. More valuable residual uncertainty and lower unit price attract capacity, while saturation limits concentration. Exponential coverage curves lead instead to logarithmic water filling, derived in Appendix E.3.

E.2. Proof of Proposition 11

Proof. The feasible region $\{F \geq 0 : \sum_k \gamma_k F_k \leq B\}$ is nonempty, convex, and compact. A maximizer exists by continuity. Positive weights and strict concavity of each separate p_k make the sum strictly concave, so the maximizer is unique.

The budget binds: otherwise some coordinate could be increased feasibly and improve the objective. Since $B > 0$, a strictly feasible point with every coordinate positive exists. The concave problem therefore satisfies the Karush–Kuhn–Tucker conditions, which are also sufficient. With budget multiplier $\Lambda \geq 0$ and nonnegativity multipliers $\tau_k \geq 0$, stationarity and complementary slackness give

$$w_k p_k'(F_k) - \Lambda \gamma_k + \tau_k = 0, \qquad \tau_k F_k = 0.$$

A funded coordinate thus satisfies $w_k p_k'(F_k) = \Lambda \gamma_k$, and an unfunded one satisfies $w_k p_k'(0) \leq \Lambda \gamma_k$. A differentiable strictly increasing concave function on $[0, \infty)$ has positive derivative at every finite argument: a zero derivative would preclude any subsequent increase by concavity. Since some coordinate is funded, $\Lambda > 0$.

For the shifted power law, $p_k'(F) = r d_k (1 + F)^{-r-1}$. Set $g_k = (w_k d_k / \gamma_k)^{1/(1+r)}$ and $\mu = (r/\Lambda)^{1/(1+r)}$. Funded stationarity gives $1 + F_k = \mu g_k$, whereas exclusion gives $\mu g_k \leq 1$. Combining them proves

$$F_k^\star = [\mu g_k - 1]_+.$$

The budget function $\sum_k \gamma_k [\mu g_k - 1]_+$ is continuous, zero for sufficiently small μ, strictly increasing once positive, and unbounded above. Exactly one μ therefore gives $B > 0$. □

For the funded set $\mathcal{K}_+ = \{k : \mu g_k > 1\}$,

$$\mu = \frac{B + \sum_{k\in\mathcal{K}_+} \gamma_k}{\sum_{k\in\mathcal{K}_+} \gamma_k g_k}. \tag{37}$$

The funded set is an upper level set of g_k, found by sorting and checking the corresponding sets. Equivalently, start with all coordinates, solve the displayed budget equation, and drop a smallest-g coordinate when $\mu g_k < 1$. Such a removal decreases μ: if $T = B + \sum \gamma_k$ and $U = \sum \gamma_k g_k$, then $(T - \gamma_j)/(U - \gamma_j g_j) < T/U$ precisely when $(T/U)g_j < 1$. A coordinate with equality has zero allocation and can be excluded without changing the solution.

E.3. Exponential coverage curves

Suppose

$$1 - p_k(F) = d_k e^{-\vartheta_k F}, \qquad d_k \in (0, 1], \quad \vartheta_k > 0.$$

Then $p_k'(F) = d_k\vartheta_k e^{-\vartheta_k F}$. The same KKT conditions give

$$F_k^\star = \left[\frac{1}{\vartheta_k} \log \frac{w_k d_k \vartheta_k}{\Lambda\gamma_k}\right]_+. \tag{38}$$

The multiplier makes the budget bind. For a known funded set $\mathcal{K}_+$,

$$\log\Lambda = \frac{\displaystyle\sum_{k\in\mathcal{K}_+} \frac{\gamma_k}{\vartheta_k} \log \frac{w_k d_k \vartheta_k}{\gamma_k} - B}{\displaystyle\sum_{k\in\mathcal{K}_+} \frac{\gamma_k}{\vartheta_k}}.$$

Funded coordinates satisfy $w_k d_k \vartheta_k > \Lambda\gamma_k$, with the reverse weak inequality for the others. This logarithmic water filling is a closed form of standard marginal equalization.

E.4. A tier-weight specialization

Consider $K + 1$ positions indexed $0, \ldots, K$, with common $d_k = d$, common $\gamma_k = \gamma$, and weights

$$w_k = (1 - \phi)\phi^k \ (k < K), \qquad w_K = \phi^K, \qquad 0 < \phi < 1.$$

They sum to one and may represent an exogenously specified load-bearing tier distribution, assumed unaffected by the allocation F. On funded coordinates, the power-law result gives

$$1 + F_k^\star \propto (1-\phi)^{1/(1+r)}\phi^{k/(1+r)} \ (k < K), \qquad 1 + F_K^\star \propto \phi^{K/(1+r)}.$$

When both final coordinates are funded,

$$\frac{1 + F_K^\star}{1 + F_{K-1}^\star} = \left(\frac{\phi}{1-\phi}\right)^{1/(1+r)}.$$

These relations concern $1 + F$, not F; unshifted ratios are approximated only at large allocations.

Funded coordinates are ranked by weights, not necessarily tier indices. With $K = 2$ and $\phi = 0.7$, weights are $(0.30, 0.21, 0.49)$. If $d = \gamma = r = 1$ and $B = 0.1$, the KKT solution is $(F_0, F_1, F_2) = (0, 0, 0.1)$ because the funded marginal $0.49/1.1^2$ exceeds 0.30 and 0.21. A small budget can therefore fund the final tier first. This is a model of simultaneous allocation across positions, not a derivation for an arbitrary sequential cascade whose tier-reach probabilities change with allocated effort.

E.5. Connecting a hard-budget subproblem to efficiency

Proposition 11 maximizes coverage value under a capacity constraint. It does not imply that an efficiency-optimal policy exhausts a computation budget. To connect the weights to J_λ, let family k have frequency ν_k, fixed hit/miss values, and fallback cost. Its benefit from increasing the hit probability has weight

$$w_k = \nu_k(v_{h,k} - v_{m,k} + \lambda b_k).$$

Positive-margin families enter the concave coverage problem. If effort F_k also adds $\rho_k F_k$ to expected per-epoch resource, the appropriate objective, up to an additive constant, is

$$\sum_k w_k p_k(F_k) - \lambda \sum_k \rho_k F_k, \qquad \sum_k \gamma_k F_k \le B.$$

The funded KKT condition becomes

$$w_k p_k'(F_k) = \lambda \rho_k + \Lambda \gamma_k,$$

with the corresponding weak inequality at zero. The budget need not bind in this penalized problem. The hard-budget result is recovered when allocation resource is separately constrained and has no additional variable charge in this objective, or when that charge is fixed across compared allocations.

E.6. Static coverage and adaptive selection

For $S \subseteq T$ and $m \notin T$,

$$\begin{aligned} p(S \cup \{m\}) - p(S) &= \mathbb{P}\left(E_m \cap \bigcap_{i \in S} E_i^c\right) \\ &\ge \mathbb{P}\left(E_m \cap \bigcap_{i \in T} E_i^c\right) = p(T \cup \{m\}) - p(T). \end{aligned}$$

This proves diminishing returns under every fixed joint law. Nonnegative marginals also prove monotonicity. The classical static cardinality result consequently applies (Nemhauser et al., 1978); a price budget instead requires an appropriate submodular-knapsack method (Sviridenko, 2004).

Static submodularity does not establish adaptive submodularity. Let exactly one of two sources succeed, each with probability $1/2$. Before observations, the second source's marginal success probability is $1/2$. After the first source fails, the conditional marginal becomes one. Information has increased its marginal benefit, violating adaptive diminishing returns. An adaptive guarantee requires the additional property studied by Golovin & Krause (2011). Likewise, subtracting modular call cost preserves submodularity but need not preserve monotonicity, so the monotone-coverage approximation guarantees do not apply unchanged to the penalized objective.

F. Portfolio complementarity

F.1. Proof of Theorem 2

Fix a portfolio S and let

$$p = p(S), \qquad \mathcal{V} = v_m + p(v_h - v_m), \qquad D = a(S) + (1-p)b > 0.$$

For a candidate $m \notin S$, write

$$r = r_m(S) = \mathbb{P}\left(E_m \cap \bigcap_{i \in S} E_i^c\right), \qquad d = d_m(S) = a(S \cup \{m\}) - a(S).$$

The enlarged portfolio has expected value $\mathcal{V} + r(v_h - v_m)$ and expected cost $D + d - rb$. Assuming this cost is positive, subtraction gives the exact identity

$$\eta(S \cup \{m\}) - \eta(S) = \frac{r\{v_h - v_m + \eta(S)b\} - \eta(S)d}{D + d - rb}. \tag{39}$$

The sign of the numerator proves the assertion. In particular, the contribution of a candidate is its success probability on the current portfolio's failure event, weighted by the delivered-value difference and the avoided fallback cost. Any change in proposal, aggregation, or verification cost belongs in d.

This calculation requires no independence among success events. It does require the same conditional values v_h, v_m and fallback cost b for the two portfolios. If a new portfolio changes which kinds of tasks reach the fallback, those conditional quantities may change; equation (39) must then be replaced by the corresponding comparison of unconditional expected values and costs.

For completeness, static coverage is monotone submodular under any fixed joint law of the events. Indeed, for $S \subseteq T$ and $m \notin T$,

$$E_m \cap \bigcap_{i \in T} E_i^c \;\subseteq\; E_m \cap \bigcap_{i \in S} E_i^c,$$

so $r_m(T) \le r_m(S)$, and every such marginal is nonnegative. This is a statement about selecting a set before observing outcomes. Conditional marginals after observing outcomes need not satisfy the analogous adaptive property.

F.2. Proof of Proposition 3

Let E_0 be the success event of the mandatory primary m_0, with probability $1/2$, and let E_1, E_2 be the success events of candidates m_1, m_2. In the first world set $E_1 = E_0$ and $E_2 = E_0^c$; in the second set $E_1 = E_0^c$ and $E_2 = E_0$. All three individual success probabilities are $1/2$ in both worlds, and all candidate costs are c. A selector whose only inputs are those marginal probabilities and costs must therefore use the same selection distribution in both worlds. Let q be its probability of choosing m_1 as the mandatory primary's partner.

In the first world, selecting m_1 gives a duplicate of the primary and requires fallback with probability $1/2$, whereas selecting m_2 covers every outcome. Thus its expected cost is $2c + qb/2$. In the second world it is $2c + (1-q)b/2$. Consequently,

$$\max\left\{2c + \frac{qb}{2},\, 2c + \frac{(1-q)b}{2}\right\} \ge 2c + \frac{b}{4}.$$

The best informed portfolio pairs the primary with its complement and costs exactly $2c$ in either world. The perfect fallback makes every policy's delivered value equal to one, so the difference cannot be explained by a quality trade-off. This establishes the proposition, including randomized selectors. The portfolio is chosen before the primary's realized output; the result does not concern a policy that learns dependence through additional observations.

F.3. Proof of Proposition ??

Let $\mathcal{T}$ be finite. Conditional on the observable state, the accepted type $T^\star \sim q$, the primary proposal $X_1 \sim p_1$, and the unsteered secondary proposal $X_2 \sim p_2$ are mutually independent. Write $a_1 = \sum_t q(t)p_1(t) < 1$. Bayes' rule gives the residual type law

$$q^-(t) = \mathbb{P}(T^\star = t \mid X_1 \neq T^\star) = \frac{q(t)(1-p_1(t))}{1-a_1}.$$

Fix a covered type set T_0, determined before the primary's proposal is realized, and set $T_1 = \mathcal{T} \setminus T_0$. Let

$$W_i = \sum_{t \in T_i} p_2(t) > 0, \qquad Z(\rho) = \rho W_0 + W_1,$$

and let p_2^ρ multiply the mass in T_0 by $\rho \in [0,1]$ and renormalize. For any real function f on $\mathcal{T}$, define

$$\overline{f}_i = \frac{1}{W_i}\sum_{t \in T_i} f(t)p_2(t), \qquad M_f(\rho) = \sum_t f(t)p_2^\rho(t).$$

Direct differentiation yields

$$M_f(\rho) = \frac{\rho W_0 \overline{f}_0 + W_1 \overline{f}_1}{Z(\rho)},$$
$$M_f'(\rho) = \frac{W_0 W_1}{Z(\rho)^2}(\overline{f}_0 - \overline{f}_1).$$

Taking $f = q^-$ proves the derivative formula for residual coverage $h(\rho)$; taking $f = q$ proves the formula for secondary standalone success probability $a_2(\rho)$. The signs do not depend on ρ. The exact verifier selects a successful candidate whenever one exists, so

$$p_{\text{pair}}(\rho) = a_1 + (1-a_1)h(\rho).$$

This proves the success-probability claim. With primary cost a, fallback cost b, and values fixed across the compared steering settings,

$$\frac{d}{dp}\frac{v_m + p(v_h - v_m)}{a + (1-p)b} = \frac{v_h(a+b) - v_m a}{[a + (1-p)b]^2}.$$

Thus increased coverage improves efficiency precisely when this numerator is positive; a weak inequality gives weak monotonicity. The independence assumptions above provide one explicit mechanism for altering residual coverage. The marginal criterion in Theorem 2 itself does not need them.

F.4. An exact four-type construction

The following finite construction separates individual quality from complementarity without estimation error. Take

$$q = (0.30, 0.20, 0.40, 0.10), \qquad p_1 = (0.45, 0.45, 0.05, 0.05),$$
$$p_2 = (0.40, 0.40, 0.10, 0.10), \qquad T_0 = \{1, 2\}.$$

Both original members have quality $1/4$. Here $W_0 = 4/5$, $W_1 = 1/5$, and

$$\overline{q}_0 = \overline{q}_1 = \frac{1}{4}, \qquad a_2(\rho) = \frac{0.20\rho + 0.05}{0.80\rho + 0.20} = \frac{1}{4}.$$

Hence steering preserves the secondary's individual quality exactly. The unnormalized residual masses are

$$q(t)(1 - p_1(t)) = (0.165, 0.110, 0.380, 0.095),$$

which sum to $3/4$. Consequently,

$$\overline{q^{-}}_0 = \frac{11}{60} < \frac{19}{60} = \overline{q^{-}}_1, \qquad h(\rho) = \frac{0.110\rho + 0.0475}{0.60\rho + 0.15}.$$

Steering away from T_0 therefore strictly increases pair coverage.

Let each primary member cost one, let fallback cost ten, and take $v_h = 1$, $v_m = 4/5$, held fixed across all settings. Thus

$$\eta(\rho) = \frac{\frac{4}{5} + \frac{1}{5}p_{\text{pair}}(\rho)}{12 - 10p_{\text{pair}}(\rho)}.$$

All entries below follow by substitution; they are exact finite-model calculations.

ρ	p_2^ρ	a_2	$h(\rho)$	p_{pair}	η
1	(.4, .4, .1, .1)	1/4	21/100	163/400	1763/15850
1/4	(.25, .25, .25, .25)	1/4	1/4	7/16	71/610
0	(0, 0, .5, .5)	1/4	19/60	39/80	359/2850

Initially, the probability of joint failure is $237/400 > 9/16$, the product of the two marginal failure probabilities. At $\rho = 1/4$, it is $9/16$; at $\rho = 0$, it is $41/80 < 9/16$. Standalone qualities are identical at all three settings.

For comparison, consider $p_3 = (0.60, 0.40, 0, 0)$. Its individual quality is $13/50 > 1/4$, yet its residual coverage is $143/750$ and its pair success with p_1 is $393/1000$, smaller than even the unsteered pair's $163/400$. Its efficiency is $4393/40350$, likewise smaller. Thus a better individual member can be a worse partner.

One can also check the explicit selection score

$$\mathrm{Sel}_\alpha(P) = (1-\alpha)\sum_{i \in P} a_i + \alpha p(P), \qquad 0 \le \alpha \le 1.$$

The score of the fully steered pair minus that of $\{1, 3\}$ is $-0.01 + 0.1045\alpha$, positive exactly when $\alpha > 20/209$. This parameterization is specific to the example, not a calibrated setting for any deployed router.

The equality of marginal quality in this construction is deliberate: in general steering can change both standalone success and residual coverage. Also, because proposals here are independent of $T^\star$ conditional on the state, an individual's quality cannot exceed $\max_t q(t) = 0.4$. The example establishes an information limitation of marginal ranking within this pool; it does not establish that steering dominates every possible additional model.

To obtain the type-dependent fallback comparison in Figure 1, keep the proposal laws and costs above, but let fallback succeed exactly on types 3 and 4. On these types the final output is always successful; on types 1 and 2 it succeeds only if the primary pair succeeds. Its unconditional delivered value is consequently

$$v(\rho) = \frac{1}{2} + \frac{1}{2}\left[1 - 0.55\left(1 - \frac{0.4\rho}{Z(\rho)}\right)\right] = 0.725 + \frac{0.11\rho}{Z(\rho)}.$$

The corresponding efficiency is $v(\rho)/[12 - 10p_{\mathrm{pair}}(\rho)]$. At $\rho = 1, 1/4, 0$ it is, respectively, $167/1585$, $127/1220$, and $29/285$: it decreases while pair coverage increases. Here the fallback's conditional value changes with the residual type distribution, so the fixed-v_m assumption does not hold. This example shows why that assumption is necessary for converting improved coverage into improved efficiency.

G. Information-state dynamics

G.1. Proof of Theorem 7

Index the first decision by $n = 0$. Let $M_{j,n} \in \{\mathrm{U}, \mathrm{I}, \mathrm{W}\}$ be the record state for cell j at the start of epoch n. The arriving cell is drawn independently of this memory. After the decision, expiration occurs and then the arrival may generate a replacing write. Write $r_j = \omega_j + \psi_j > 0$ and

$$\alpha_j = (1 - \delta_j)(1 - \nu_j r_j) < 1.$$

An informative record is present at the next epoch if an informative write occurs, or if no write occurs and the existing informative record survives expiration. These two events are disjoint. Therefore

$$i_{j,n+1} = \nu_j \omega_j + \alpha_j i_{j,n}, \qquad\qquad w_{j,n+1} = \nu_j \psi_j + \alpha_j w_{j,n}. \tag{40}$$

Their fixed points and empty-start solutions are

$$\begin{aligned} i_j^* &= \frac{\nu_j \omega_j}{1 - \alpha_j}, & \qquad w_j^* &= \frac{\nu_j \psi_j}{1 - \alpha_j}, \\ i_{j,n} &= i_j^*(1 - \alpha_j^n), & \qquad w_{j,n} &= w_j^*(1 - \alpha_j^n). \end{aligned}$$

For other initial states, subtracting the fixed-point equations gives $i_{j,n} - i_j^* = \alpha_j^n(i_{j,0} - i_j^*)$, and the analogous formula for w.

Let $\bar{v}_{\mathrm{U}} = \sum_j \nu_j v_j^{\mathrm{U}}$ and $\bar{c}_{\mathrm{U}} = \sum_j \nu_j c_j^{\mathrm{U}} > 0$, so $\eta_{\mathrm{U}} = \bar{v}_{\mathrm{U}}/\bar{c}_{\mathrm{U}}$. Define

$$g_j^{\mathrm{I}} = v_j^{\mathrm{I}} - v_j^{\mathrm{U}} - \eta_{\mathrm{U}}(c_j^{\mathrm{I}} - c_j^{\mathrm{U}}), \qquad g_j^{\mathrm{W}} = v_j^{\mathrm{W}} - v_j^{\mathrm{U}} - \eta_{\mathrm{U}}(c_j^{\mathrm{W}} - c_j^{\mathrm{U}}).$$

Since the current arrival is independent of the pre-decision memory,

$$\begin{aligned} \mathbb{E}[V_n - \eta_{\mathrm{U}} D_n] &= \sum_j \nu_j \left[v_j^{\mathrm{U}} - \eta_{\mathrm{U}} c_j^{\mathrm{U}} + i_{j,n} g_j^{\mathrm{I}} + w_{j,n} g_j^{\mathrm{W}}\right] \\ &= \sum_j \nu_j G_j (1 - \alpha_j^n), \end{aligned} \tag{41}$$

where

$$G_j = i_j^* g_j^{\mathrm{I}} + w_j^* g_j^{\mathrm{W}} = \frac{\nu_j(\omega_j g_j^{\mathrm{I}} + \psi_j g_j^{\mathrm{W}})}{1 - \alpha_j}.$$

The cold-policy terms cancel after averaging over cells. The two appearances of ν_j have different roles: the one inside G_j determines how quickly that cell acquires records; the outer one weights how often it is used for a decision.

Let $\bar{c}_\infty > 0$ be the stationary expected cost. Passing to the limit in equation (41) gives the stronger exact comparison

$$\eta_\infty - \eta_{\mathrm{U}} = \frac{\sum_j \nu_j G_j}{\bar{c}_\infty}. \tag{42}$$

This proves the stationary sign criterion. Summing the finite-time identity instead gives

$$\sum_{n=0}^{N-1} \mathbb{E}[V_n - \eta_{\mathrm{U}} D_n] = \sum_j \nu_j G_j \left[N - \frac{1-\alpha_j^N}{1-\alpha_j}\right]. \tag{43}$$

This is an identity for aggregate expected surplus, not an interchange of expectation with a random value-to-cost ratio.

If $g_j^{\mathrm{I}} > 0 > g_j^{\mathrm{W}}$, the sign of G_j is the sign of

$$\omega_j g_j^{\mathrm{I}} + \psi_j g_j^{\mathrm{W}} = r_j \left[g_j^{\mathrm{I}} - \frac{\psi_j}{r_j}(g_j^{\mathrm{I}} - g_j^{\mathrm{W}})\right].$$

It is positive exactly when $\psi_j / r_j < g_j^{\mathrm{I}}/(g_j^{\mathrm{I}} - g_j^{\mathrm{W}})$.

For finitely many cells, every marginal mean formed from fixed conditional cell values converges geometrically, by equation (40), with factor at most $\max_j \alpha_j < 1$. The same holds for the ratio of expected value to expected cost once the latter is bounded away from zero. These statements do not require the hit sequence itself to be independent or Markov. In fact, persistent memory generally correlates successive hits even when arrivals are independent.

G.2. An arbitrary reference and memory-maintenance overhead

The cold state of a memory-enabled controller can already incur a maintenance or logging bill. To compare with a reference that does not pay that bill, fix any $\lambda \in \mathbb{R}$ and define

$$\begin{aligned} J_U(\lambda) &= \sum_j \nu_j (v_j^U - \lambda c_j^U), \\ g_j^s(\lambda) &= (v_j^s - v_j^U) - \lambda(c_j^s - c_j^U), \qquad s \in \{I, W\}, \\ G_j(\lambda) &= \frac{\nu_j[\omega_j g_j^I(\lambda) + \psi_j g_j^W(\lambda)]}{1-\alpha_j}. \end{aligned}$$

The exact empty-start finite-horizon identity is

$$\sum_{n=0}^{N-1} \mathbb{E}[V_n - \lambda D_n] = N J_U(\lambda) + \sum_j \nu_j G_j(\lambda) \left[N - \frac{1-\alpha_j^N}{1-\alpha_j}\right]. \tag{44}$$

At stationarity the expected surplus is $J_U(\lambda) + \sum_j \nu_j G_j(\lambda)$; with positive stationary expected cost, its sign is the sign of $\eta_\infty - \lambda$.

In particular, let a no-memory reference have the same mean value as the cold state, mean cost $c_0 > 0$, and efficiency η_0. If the memory-enabled cold state costs $c_0 + h$ with $h > 0$, then $J_U(\eta_0) = -\eta_0 h$. Memory is therefore at least as efficient as that reference exactly when

$$\sum_j \nu_j G_j(\eta_0) \geq \eta_0 h. \tag{45}$$

Here h is the difference in expected cost per epoch; any further record-dependent costs remain inside $g_j^s(\eta_0)$.

To prove these formulas, condition on the arriving cell and its record state as in equation (41), retaining the cold term instead of canceling it. The expected surplus at epoch n is $J_U(\lambda) + \sum_j \nu_j G_j(\lambda)(1-\alpha_j^n)$. Summing the geometric series gives equation (44), and taking the limit gives the stationary statement. Finally, the no-memory reference value equals $\eta_0 c_0$, so its comparison with cold cost $c_0 + h$ gives $J_U(\eta_0) = -\eta_0 h$ and equation (45).

G.3. Exact coverage for time-based and event-based retention

Consider first a record that remains usable for exactly E global epochs after it is written, with no other expiration. Assume E is a positive integer. Once the initial transient has passed, cell j is covered precisely when at least one of the preceding E epochs wrote a record for it. The iid write probability per epoch is $\nu_j r_j$, and hence

$$\chi_j^{\mathrm{TTL}} = 1 - (1-\nu_j r_j)^E, \qquad i_j^{\mathrm{TTL}} = \frac{\omega_j}{r_j}\chi_j^{\mathrm{TTL}}, \qquad w_j^{\mathrm{TTL}} = \frac{\psi_j}{r_j}\chi_j^{\mathrm{TTL}}. \tag{46}$$

Conditional on a write to j, its type is informative with probability ω_j/r_j. This remains true for the most recent such write, which proves the last two identities.

Now consider an ordinary FIFO that retains the most recent M write events globally, including repeated entries for the same cell, and answers a query using the newest retained matching record. Put

$$\beta = \sum_j \nu_j r_j > 0, \qquad q_j = \frac{\nu_j r_j}{\beta}.$$

Discarding epochs with no write produces an iid sequence of cell marks with law q. Thus, after the FIFO has filled,

$$\chi_j^{\mathrm{FIFO}} = 1 - (1-q_j)^M, \qquad i_j^{\mathrm{FIFO}} = \frac{\omega_j}{r_j}\chi_j^{\mathrm{FIFO}}, \qquad w_j^{\mathrm{FIFO}} = \frac{\psi_j}{r_j}\chi_j^{\mathrm{FIFO}}. \tag{47}$$

Although M write events span M/β global epochs on average, replacing E by that average in equation (46) is not exact: coverage is a nonlinear function of the random window length. A cache that deduplicates cells or uses a different replacement policy is a different model.

G.4. Oracle regret and its signed transient

The next identity concerns scalar utility, not necessarily raw accuracy. Let $s_j^{\mathrm{U}}, s_j^{\mathrm{I}}, s_j^{\mathrm{W}}$ be fixed conditional means, and let s_j^{O} denote a per-cell oracle mean. For example, $s = v - \lambda c$ gives the additive objective. Assume no expiration, empty initial memory, and finitely many cells. Put

$$s_j^{\infty} = \frac{\omega_j s_j^{\mathrm{I}} + \psi_j s_j^{\mathrm{W}}}{r_j}.$$

The probability that no write to j has occurred by epoch n is $(1-\nu_j r_j)^n$. Conditional on any write having occurred, the newest record has the informative/wrong mixture used in s_j^{∞}. Consequently,

$$\mathbb{E}[s_{J_n}^{\mathrm{O}} - s_n] = \sum_j \nu_j \left[s_j^{\mathrm{O}} - s_j^{\infty} + (1-\nu_j r_j)^n (s_j^{\infty} - s_j^{\mathrm{U}})\right].$$

Summing the geometric series gives

$$\begin{aligned}\mathbb{E}[\mathrm{Reg}_N] = N \sum_j \nu_j (s_j^{\mathrm{O}} - s_j^{\infty}) \\ + \sum_j \frac{1-(1-\nu_j r_j)^N}{r_j}(s_j^{\infty} - s_j^{\mathrm{U}}).\end{aligned} \tag{48}$$

The linear term may be split as

$$s_j^{\mathrm{O}} - s_j^{\infty} = (s_j^{\mathrm{O}} - s_j^{\mathrm{I}}) + \frac{\psi_j}{r_j}(s_j^{\mathrm{I}} - s_j^{\mathrm{W}}),$$

separating the limitation of informative records from the effect of incorrect records. The transient term in equation (48) is signed. Its valid general bound is

$$\left|\sum_j \frac{1-(1-\nu_j r_j)^N}{r_j}(s_j^{\infty} - s_j^{\mathrm{U}})\right| \le \sum_j \frac{|s_j^{\infty} - s_j^{\mathrm{U}}|}{r_j}.$$

If every $s_j^{\infty} \ge s_j^{\mathrm{U}}$, the transient is nonnegative, increases with N, and converges to $\sum_j (s_j^{\infty} - s_j^{\mathrm{U}})/r_j$. Without that condition, interpreting it as a positive learning cost is inappropriate: the limiting record policy may be worse than the cold policy.

G.5. Proof of Theorem 8

Let $\kappa_z = 1/\zeta(z)$ and $\nu_j = \kappa_z j^{-z}$ for $j \geq 1$, where $z > 1$. Assume clean writes with a common probability $\omega > 0$ and a constant positive informed-minus-cold hit gap Δ. Equation (46) makes the TTL coverage deficit

$$L_E = \Delta \sum_{j \geq 1} \nu_j (1 - \omega \nu_j)^E.$$

For $0 \leq x < 1$, $(1-x)^E \leq e^{-Ex}$; for $0 \leq x \leq 1/2$, it is also at least e^{-2Ex}. Only finitely many cells can have $\omega\nu_j > 1/2$, and each of their contributions vanishes exponentially. It therefore suffices to establish

$$\sum_{j \geq 1} \kappa_z j^{-z} \exp(-c\kappa_z j^{-z}) = \Theta(c^{1/z-1}), \qquad c \longrightarrow \infty. \tag{49}$$

For $f_c(x) = \kappa_z x^{-z} e^{-c\kappa_z x^{-z}}$, substitution $x = (c\kappa_z)^{1/z} u$ yields

$$\int_1^\infty f_c(x)\, dx = \kappa_z^{1/z} c^{1/z-1} \int_{(c\kappa_z)^{-1/z}}^\infty u^{-z} e^{-u^{-z}}\, du.$$

The integral on the right tends to the finite positive constant

$$\int_0^\infty u^{-z} e^{-u^{-z}}\, du = \frac{1}{z}\Gamma\left(1 - \frac{1}{z}\right).$$

The function f_c is unimodal. Splitting its increasing and decreasing parts gives a sum–integral discrepancy bounded by a constant times its maximum, which is at most $1/(ce)$. This is smaller than $c^{1/z-1}$. Thus equation (49) holds. Applying it with $c = \omega E$ and $2\omega E$ proves

$$L_E = \Theta\left(E^{-(z-1)/z}\right).$$

With no expiration and empty start, the uncovered probability at time n is instead $(1-\omega\nu_j)^n$, so its per-epoch information deficit is L_n for $n \geq 1$. Summing this power law, with exponent $(z-1)/z \in (0,1)$, gives

$$\sum_{n=0}^{N-1} \Delta \sum_j \nu_j (1-\omega\nu_j)^n = \Theta(N^{1/z}).$$

Equivalently, this sum equals

$$\frac{\Delta}{\omega} \sum_{j \geq 1} \left[1 - (1-\omega\nu_j)^N\right].$$

The result describes missing-information loss. Any persistent informed-versus-oracle gap contributes a separate linear term. The common positive gap and write-rate assumptions matter; Zipf arrivals alone do not prescribe the rate when those quantities vary freely with the cell.

The leading constants follow from a sharper squeeze. For any fixed $\varepsilon > 0$, sufficiently small $x \geq 0$ satisfies $e^{-(1+\varepsilon)nx} \leq (1-x)^n \leq e^{-nx}$; the finitely many remaining cells contribute only exponentially small terms. Applying the integral equivalent above and then taking $\varepsilon \downarrow 0$ gives

$$L_E \sim \frac{\Delta \kappa_z^{1/z}}{z} \Gamma\left(1 - \frac{1}{z}\right) (\omega E)^{1/z-1}.$$

Summing this equivalent for the no-expiration process gives the cumulative uncovered deficit

$$R_N := \sum_{n=0}^{N-1} L_n \sim \Delta \kappa_z^{1/z} \Gamma\left(1 - \frac{1}{z}\right) \omega^{1/z-1} N^{1/z},$$

where the initial term is $L_0 = \Delta$. These are the asymptotic reference curves used in the numerical illustrations.

G.6. What arm coverage does, and does not, establish

Cell coverage is not equivalent to identifying the best allocation within a cell. The distinction can be made explicit through an idealized record model.

Lemma 12 (Exact-reveal arm coverage)**.** *Fix a cell and a finite set of arms with values μ_m. On each visit, an independent logger evaluates arm m with probability e_m. A logged evaluation reveals μ_m exactly. The deployed recommendation chooses a highest-valued recorded arm, using an arbitrary initial action if no arm has been recorded. If a best arm m^* satisfies $e_{m^*} > 0$, then after v visits the probability of a suboptimal recommendation is at most $(1 - e_{m^*})^v$. If no best arm is in the logger's support, the recommendation converges to the best value within that support.*

Proof. The probability that m^* has never been recorded is $(1 - e_{m^*})^v$. Once it is recorded, its exact value is available and no recorded arm has greater value. This proves the first assertion. For the second, each arm with positive logging probability is eventually recorded almost surely; finiteness of the arm set implies that eventually all such arms have been recorded. The recommendation then maximizes over precisely that set. □

The rule in this lemma compares recorded arms. An unrecorded arm with an optimistic fixed prior could otherwise outrank a discovered best arm forever. The logger is an explicit source of information, not a consequence of greedy recommendations. If logging uses auxiliary evaluations, their cost must be included in allocation overhead; if logging replaces deployed actions, its exploration losses also count.

For noisy evaluations, the exact-reveal assumption can be replaced by an ordinary estimation model, but the resulting statement is different. Suppose independent observations of arm m lie in $[0, 1]$ with mean μ_m, and logging is independent of these observations. Let $\widehat{\mu}_m$ be its sample mean, and recommend the best sampled arm. Assume a unique best arm m^* with positive logging probability, and let $\Delta_m = \mu_{m^*} - \mu_m > 0$ for each suboptimal arm in the logger's support. Conditional on positive counts N_{m^*} and N_m, Hoeffding's inequality and a union bound give

$$\mathbb{P}(\widehat{\mu}_m \geq \widehat{\mu}_{m^*} \mid N_{m^*}, N_m) \leq e^{-N_{m^*}\Delta_m^2/2} + e^{-N_m\Delta_m^2/2}.$$

Since $N_m \sim \mathrm{Binomial}(v, e_m)$, a valid unconditional upper bound on the probability of a suboptimal recommendation is

$$(1 - e_{m^*})^v + \sum_{\substack{m \neq m^* \\ e_m > 0}} \left[\left(1 - e_{m^*} + e_{m^*} e^{-\Delta_m^2/2}\right)^v + \left(1 - e_m + e_m e^{-\Delta_m^2/2}\right)^v \right].$$

To see this, first account for the event that the best arm has not been sampled, and then apply the conditional bound to each sampled suboptimal arm; allowing zero counts in the exponential moments only enlarges the bound. Unlike the exact-reveal lemma, this rate depends on quality gaps and sampling noise. Persistent deployment exploration may still incur linear cumulative cost even when the recommendation becomes accurate.

Finally, binary verifier noise can alter the estimand. With arm-independent false-accept and false-reject probabilities β_{fa} and β_{fr}, an arm's recorded success mean is

$$\widetilde{\mu}_m = \beta_{\mathrm{fa}} + (1 - \beta_{\mathrm{fr}} - \beta_{\mathrm{fa}})\mu_m.$$

Rankings are preserved when $\beta_{\mathrm{fr}} + \beta_{\mathrm{fa}} < 1$. Arm-dependent errors need not preserve rankings, regardless of sample size. Thus adequate coverage, accurate mean estimation, and faithful verification are distinct requirements.

H. A barrier-round extension

The portfolios in the main text use OR semantics: one successful candidate is sufficient. The following separate extension instead uses AND semantics: b independent decisions share a barrier, and any unresolved decision causes an additional wait. It describes two fixed fallback policies, not an optimal scheduling rule over arbitrary adaptive policies.

Let $L > 0$ be the deterministic primary-round latency. Policy $X \in \{\mathrm{s}, \mathrm{f}\}$ gives each member a stationary primary-hit probability $\pi_X \in (0, 1)$, a final expected value $V_X > 0$, and an additional round latency $\Delta_X \geq 0$ whenever at least one member misses. Under independence of member hits within a stationary round, the expected latency is $L + (1 - \pi_X^b)\Delta_X$. Its value per member per unit round latency is therefore

$$\eta_X(b) = \frac{V_X}{L + (1 - \pi_X^b)\Delta_X}.$$

The hit rates may differ because the fallback policies leave different subsequent states. For example, for independent two-state member chains with hit-conditional transition probability p_{h} and miss-conditional transition probability $p_{\mathrm{m},X}$, $\pi_X = p_{\mathrm{m},X}/(1 - p_{\mathrm{h}} + p_{\mathrm{m},X})$.

Proposition 13 (Mixed fallback crossing)**.** *Assume $V_{\mathrm{s}} > V_{\mathrm{f}} > 0$, $\Delta_{\mathrm{s}} > \Delta_{\mathrm{f}} \geq 0$, and $0 < \pi_{\mathrm{f}} < \pi_{\mathrm{s}} < 1$. On the continuous range $b > 0$, there is exactly one finite positive crossing between the two efficiencies if and only if*

$$\frac{V_{\mathrm{s}}}{L + \Delta_{\mathrm{s}}} < \frac{V_{\mathrm{f}}}{L + \Delta_{\mathrm{f}}}.$$

When it exists, the slow policy is better before the crossing and the fast policy afterwards. Otherwise the slow policy is better at every finite positive b.

Proof. Cross-multiplying positive denominators shows that the sign of $\eta_{\mathrm{s}}(b) - \eta_{\mathrm{f}}(b)$ is the sign of

$$\Psi(b) = \Psi_\infty + V_{\mathrm{f}}\Delta_{\mathrm{s}}\pi_{\mathrm{s}}^b - V_{\mathrm{s}}\Delta_{\mathrm{f}}\pi_{\mathrm{f}}^b,$$

where $\Psi_\infty = V_{\mathrm{s}}(L + \Delta_{\mathrm{f}}) - V_{\mathrm{f}}(L + \Delta_{\mathrm{s}})$. At zero, $\Psi(0) = (V_{\mathrm{s}} - V_{\mathrm{f}})L > 0$, and its limit is Ψ_∞. If $\Delta_{\mathrm{f}} = 0$, it is strictly decreasing. Otherwise,

$$\Psi'(b) = -V_{\mathrm{f}}\Delta_{\mathrm{s}}|\log \pi_{\mathrm{s}}|\pi_{\mathrm{s}}^b + V_{\mathrm{s}}\Delta_{\mathrm{f}}|\log \pi_{\mathrm{f}}|\pi_{\mathrm{f}}^b.$$

The ratio of the positive term to the magnitude of the negative term is a positive constant times $(\pi_{\mathrm{f}}/\pi_{\mathrm{s}})^b$, which strictly decreases to zero. Thus the derivative can change sign at most once, from positive to negative. The function either decreases throughout, or first increases and then decreases, with no interior minimum. Starting from a positive value, it crosses zero exactly once if and only if $\Psi_\infty < 0$, proving the statement. $\square$

When the fast policy has no extra latency, set $\Delta_{\mathrm{f}} = 0$ and define

$$\Xi = 1 - \frac{L}{\Delta_{\mathrm{s}}}\left(\frac{V_{\mathrm{s}}}{V_{\mathrm{f}}} - 1\right).$$

A finite positive crossing exists exactly when $0 < \Xi < 1$, and then $b^* = \log \Xi / \log \pi_{\mathrm{s}}$. For physical integer fan-outs, compare the efficiencies at the neighboring integers; when $b^* < 1$, the fast policy is already better at every admissible fan-out.

For the equal-value case $V_{\mathrm{s}} = V_{\mathrm{f}}$, comparison reduces to the excess latency

$$\Phi(b) = (\Delta_{\mathrm{s}} - \Delta_{\mathrm{f}}) - \Delta_{\mathrm{s}}\pi_{\mathrm{s}}^b + \Delta_{\mathrm{f}}\pi_{\mathrm{f}}^b.$$

Its derivative changes sign at most once, from negative to positive, and $\Phi(0) = 0$, $\Phi(\infty) > 0$. Hence on $b \geq 1$, either $\Phi(1) < 0$ and there is one later crossing, or $\Phi(1) \geq 0$ and the fast policy is never worse. These formulas assume deterministic round latency and parallel fallback across missed members; they do not model heavy-tailed completion times or serial repairs.

I. Reproducing the Analytical Illustrations

The accompanying source contains `scripts/reproduce.py`. Running this script with Python 3, NumPy, SciPy, and Matplotlib regenerates both figures, writes their numerical coordinates as CSV tables under `numerical_results`, and checks the finite construction using rational arithmetic. The calculations use no model calls or benchmark measurements. All costs and values in the examples are illustrative model parameters.

Figure 1, left, uses the four-type construction in the preceding appendix. Every proposer costs one and the fallback costs ten. The type-independent fallback succeeds with probability 0.8 on a rejected task. The alternative fallback succeeds exactly on types 3 and 4, so its unconditional delivered value is computed as

$$p(\rho) + \sum_{t=3}^{4} q(t)(1 - p_1(t))(1 - p_2^{\rho}(t)).$$

Both variants have cost $2 + 10(1 - p(\rho))$. This calculation accounts for the changed distribution of rejected types; it does not hold the conditional fallback value fixed. The right panel evaluates the three affine cost functions in Equation (16) on a grid of sufficiency probabilities. The exact band endpoints are $1/45$ and $4/5$.

Figure 2, left, uses ten equally likely cells, $r_j = 0.5$, $\delta_j = 0.01$, unit expected cost in all record states, and values $(0.6, 0.9, 0.1)$. The retention factor is 0.9405. With erroneous fraction e, the expected value at epoch n is

$$0.6 + \frac{0.05}{1 - 0.9405}(1 - 0.9405^n)(0.3 - 0.8e).$$

This is an evaluation of the exact recurrence, not a simulated learning curve.

The Zipf panels use $\nu_j = cj^{-2}$ with $c = 1/\zeta(2)$ and clean writes at every epoch. The missing coverage and its cumulative value are

$$D_N = \sum_{j \geq 1} \nu_j (1 - \nu_j)^N, \qquad R_N = \sum_{j \geq 1} \left[1 - (1 - \nu_j)^N\right].$$

Their leading asymptotic equivalents are $D_N \sim \sqrt{c\pi}\, N^{-1/2}/2$ and $R_N \sim \sqrt{c\pi}\, N^{1/2}$, as obtained by the integral argument for Theorem 8. The script evaluates the first 200,000 cells and adds the first-order tail contribution. If $T_2 = \sum_{j>200{,}000} \nu_j^2$, the resulting overestimation is at most NT_2 for D_N and $N(N-1)T_2/2$ for R_N. Across the displayed horizons $10 \leq N \leq 10^6$, these bounds are below 1.6×10^{-11} and 7.8×10^{-6}, respectively. The bounds and plotted values are included in the CSV output.

The source also checks the distinction between time-based and event-based memory: a cell written with probability 0.1 per epoch, a total write probability 0.5, and a FIFO holding two events has coverage 0.36. Treating its mean four-epoch span as a deterministic TTL instead gives 0.3439. Finally, explicit sums of the memory recurrence are compared with Equation (24) for both positive and negative record gains.

J. Experimental Details and Full Results

J.1. Protocol

Data. CodeRouterBench (`id_results_long.csv`) gives, for 9,999 coding tasks and 8 models (claude-opus-4-6, claude-sonnet-4-6, gpt-5.4, Qwen3-Max, kimi-k2.5, qwen3.5-plus, glm-5, MiniMax-M2.7), an execution-verified score in $[0, 1]$, the billed cost in dollars, and the wall-clock latency in milliseconds. Tasks carry a dimension (9 values) and a source benchmark; the 21 non-empty dimension×source pairs are the cells of Section 5. The provider's split is probing (7,080 tasks; the original train and validation sets) and `id_test` (2,919 tasks). Cross-checks asserted in every script: always-Opus on `id_test` bills \$34.02 and scores 0.4383, as in the benchmark paper. Dollars use the price list of the benchmark's `models.json`; 148 Qwen3-Max rows are missing runs recorded with cost 0 and score 0 and are kept as recorded (sensitivities are reported below); 3,150 rows have latency 0 and are dropped from latency computations only.

Reference, verifier, accounting. The reference B is always claude-opus-4-6, so (Y_B, C_B) is observed for every task. A verified cascade attempts a primary P (one model, or a portfolio in Appendix J.3), accepts when $H = \mathbf{1}\{\text{score}_P \geq \theta\}$ with $\theta \in \{1, 0.5\}$, and commits Opus on a miss. Hence $V = HY_P + (1 - H)Y_B$, $A = C_P$, $(Y_F, C_F) = (Y_B, C_B)$, and $D = C_P + (1 - H)C_B$ in dollars; in seconds, sequential escalation adds latencies and hedged execution takes the maximum on a miss. A noisy verifier flips the exact verdict with false-accept rate β_{fa} and false-reject rate β_{fr} using one fixed uniform draw per task. Ex-ante routers are the benchmark's nine released decision files (fine-tuned Qwen routers of 0.8B–27B, logistic regression, TF–IDF MLP, RouteLLM matrix factorization and similarity-weighted, DimensionBest), trained on probing and evaluated on their `id_test` decisions; an ex-ante router pays only its chosen path.

Estimation. Prediction primitives (p, v_h, v_m, $\bar{v}$, $\bar{c}$, a, b, $r_m(S)$, W_λ, s, ν_j, ω_j, ψ_j, g_j^s, p_h, p_m, ℓ_p, ℓ_b, $p_k(F)$) are means on probing. Realized quantities are means on `id_test`. CIs are 95% percentile bootstrap intervals over task resampling (paired when two policies are compared on the same tasks): 1,000 draws in X1, X3, and the X6 evaluation; 500 in X2, X4, X5, and the X6 exponent fits. Stream experiments (X4, X5) draw tasks i.i.d. with replacement from `id_test` and add seed bootstraps over streams where the quantity is a probability. All scripts use seed 20260911.

Test kinds and verdicts. Each claim is one of: IDENTITY (algebra of the paper evaluated on the data; can fail only through implementation and is reported as a code check), IN-SAMPLE hypothesis test (a stated hypothesis measured directly on `id_test`), OUT-OF-SAMPLE prediction (probing primitives, `id_test` realization), or SEMI-SYNTHETIC (a documented construction over real outcomes). Verdicts are PASS, FAIL, INCONCLUSIVE (CI does not decide), and NOT INSTANTIABLE (the hypotheses cannot be realized on the data); PARTIAL is used where a multi-part hypothesis holds in part. Pass criteria were declared in each script before the run; the exceptions are listed in Appendix J.8.

Audit. Each script was re-run from its original version and reproduced its stored results with 0 numeric mismatches; headline numbers were recomputed independently with plain pandas code; hypotheses were re-read against the current `.tex` sources; verdict changes and corrected numbers are recorded in the README Audit sections and are the values used here. Original scripts and outputs are kept as `run_orig.py` and `results_orig.json`.

J.2. X1: Reference-relative surplus and the homogeneous cascade

Results tested and hypotheses. Theorem 1 with (3), (4), (5), (6) (no hypothesis beyond integrability and positive expected costs); (7), (8), (9) under the paper's hypothesis "a matching fallback $v_m = \bar{v}$, $b = \bar{c}$, and $v_h, \bar{v} > 0$"; the ex-ante form $(tG + 1 - t)/(ta/b + 1 - t)$ of Appendix A.2 under the same matching; the exact condition (32) and its sufficient pair $v_h \geq \bar{v}_h$, $a \leq p\bar{c}_h$ (Appendix A.3); the remark that (2) is a ratio of aggregates; and the remark of Appendix A that $Q \geq -e$ with $e = \mathbb{P}(H = 1, Y_P < Y_B)$. Configurations: 7 primaries $\times$ 2 thresholds, pooled and per dimension, exact and noisy verifiers with $(\beta_{\text{fa}}, \beta_{\text{fr}}) \in \{(0.1, 0.1), (0.3, 0.1), (0.1, 0.3)\}$, dollars and seconds; 340 in-sample, 154 out-of-sample, 140 half-split, and 9 ex-ante records.

Prediction with plugged primitives. For Qwen3-Max at $\theta = 1$ on probing, $p = 0.2144$, the floor $(\bar{v}/v_h)(a/b) = 0.0185$, and (9) gives $\eta/\eta_B = 1.550$; the realized `id_test` ratio is 1.678 $[1.585, 1.783]$, a relative error of -0.076 $[-0.135, -0.019]$ and the only under-prediction among the 14 cascades. The state-matched form $[p\, v_h + (1-p)\bar{v}_m]/[a + (1-p)\bar{c}_m]/\eta_B$ with the same probing primitives has out-of-sample relative errors within ± 0.067 on all 14 cascades.

Hypothesis violations measured. Matched fallback: $v_m/\bar{v}$ and $b/\bar{c}$ as in Table 1; $\bar{c}_h/\bar{c} \in [0.83, 1.68]$. Surplus terms on `id_test` at $\theta = 1$ (dollars per task $\times 10^3$): Qwen3-Max avoided reference work $\mathbb{E}[HC_B] = 4.17$, primary bill $\mathbb{E}[A] = 0.52$, $Q/\eta_B = 1.78$, $\eta/\eta_B = 1.678$ $[1.585, 1.783]$; claude-sonnet-4-6 avoided 2.23, bill 8.58, $Q/\eta_B = 0.49$, $\eta/\eta_B = 0.675$ $[0.659, 0.689]$. Ex-ante routers: realized $\eta/\eta_B \in [3.39, 6.32]$ against the matched formula's 2.1–4.1; with $b = \mathbb{E}[C_B \mid \text{Opus branch}]$ instead of $\bar{c}$ the error flips to median -0.097 $[-0.134, -0.066]$, CI excluding 0 in 9/9. Dropping the 148 zero-cost Qwen3-Max rows moves p from 0.2203 to 0.2235 and the floor from 0.0197 to 0.0204 at $\theta = 1$ with no sign change; the alternative price list of the benchmark paper changes the in-sample `id_test` floor sign count from 9/14 to 10/14 (out of sample, probing $\rightarrow$ `id_test`, from 8/14 to 9/14) with no verdict change.

Table 1. X1 claims. Kinds: I = identity (code check), IS = in-sample hypothesis test, OOS = out-of-sample prediction. Verdicts after audit.

	Claim	Kind	Measurement	Verdict
C1	(4): $\eta - \eta_B = (Q + \eta_B \mathcal{S})/(\bar{c} - \mathcal{S})$	I	max relative residual 5.9×10^{-16} (dollars), 5.5×10^{-15} (seconds) over 340 configurations $\times$ 1,001 bootstrap rows	PASS
C2	(6) four-term decomposition	I	residual 3.3×10^{-15}; miss-quality and recovery-cost terms exactly 0 with Opus fallback	PASS
C3	hypothesis $v_m = \bar{v}$	IS	$v_m/\bar{v} \in [0.356, 0.853]$, CI excludes 1 in 28/28 pooled and 201/252 dimension configurations; $\bar{v}_h/\bar{v} \in [1.58, 2.08]$	FAIL
C4	hypothesis $b = \bar{c}$	IS	$b/\bar{c} \in [0.813, 1.119]$, CI excludes 1 in 24/28 pooled (both directions) and 172/252 dimension configurations	FAIL (relaxed rule); INCONCLUSIVE at the pre-registered 90% rule
C5	(8) sign as stated	IS	agreement 19/28 pooled (16/20 decided), 223/252 dimension, 34/42 noisy, 10/14 at the alternative price list; all disagreements are predicted wins that lose	FAIL
C6	(32) iff $\eta \geq \eta_B$	I	residual 1.2×10^{-14}; sign 28/28, 252/252, 42/42	PASS
C7	$v_h \geq \bar{v}_h$ and $a \leq p\bar{c}_h$ imply $\eta \geq \eta_B$	I/IS	0 violations; certifies 12/16 pooled and 80/117 dimension winners	PASS
C8	(9), probing $\to$ `id_test`	OOS	median relative error $+0.153$, mean $\lvert$error$\rvert$ 0.174, CI covers 0 in 2/14, over-prediction 13/14, sign 8/14; dimension mean $\lvert$error$\rvert$ 0.263	FAIL
C9	state-matched form, probing $\to$ `id_test`	OOS	mean $\lvert$error$\rvert$ 0.027, max 0.067, CI covers 0 in 12/14; dimension 121/126; random halves 132/140	PASS (split stability of 7 primitives)
C10	ex-ante formula, matched $b = \bar{c}$, $v_m = \bar{v}$, 9 routers	IS	relative error median -0.376, range $[-0.491, -0.315]$, CI excludes 0 in 9/9; $v_m/\bar{v} \in [1.22, 1.62]$, $b/\bar{c} \in [0.52, 0.65]$	FAIL
C11	ex-ante vs verified cascade on the same decisions	I	$D_{\text{casc}} - D_{\text{ex}} = \mathbb{E}[T(1-H)C_B]$ to 7.1×10^{-15}; extra fallback \$0.00484–0.00642 per task, discarded primary bill \$0.00017–0.00129, value gain 0.021–0.049; ex-ante more efficient 9/9	PASS
C12	$\eta \neq$ mean of V_n/D_n	IS	mean$(V/D)/\eta \in [2.34, 33.2]$ (median 5.31); winner sign flips in 10/28 pooled and 70/252 dimension configurations	PASS (descriptive)
C13	$Q \geq -e$; $\eta_B \mathcal{S} \geq e$ suffices	I	holds 340/340; noisy verifiers give $Q < 0$ in 2/42 (min -0.0046)	PASS

Table 2. X1: the 14 verified cascades on `id_test` (exact verifier, Opus fallback, CSV prices). Floor is $(\bar{v}/v_h)(a/b)$ with $b = \bar{c}$; "sign" is whether sign$(p - \text{floor})$ equals sign$(\eta - \eta_B)$; the last column is the in-sample relative error of (9). Alternative reading $b = \bar{c}_m$ (realized fallback bill on misses): sign agreement 24/28 pooled, 216/252 dimension, but larger value error (mean $\lvert$error$\rvert$ 0.198 in-sample, 0.227 out of sample).

Primary	θ	p	floor	η/η_B	95% CI	sign	rel. err. (9)
Qwen3-Max	1.0	0.220	0.020	1.68	[1.58, 1.78]	yes	-0.074
Qwen3-Max	0.5	0.369	0.022	2.00	[1.89, 2.14]	yes	$+0.024$
kimi-k2.5	1.0	0.166	0.030	1.23	[1.19, 1.27]	yes	$+0.091$
kimi-k2.5	0.5	0.343	0.035	1.41	[1.35, 1.47]	yes	$+0.312$
qwen3.5-plus	1.0	0.210	0.064	1.35	[1.29, 1.42]	yes	$+0.003$
qwen3.5-plus	0.5	0.350	0.071	1.51	[1.44, 1.58]	yes	$+0.146$
glm-5	1.0	0.217	0.215	0.91	[0.87, 0.94]	no	$+0.103$
glm-5	0.5	0.370	0.239	0.98	[0.94, 1.01]	no	$+0.269$
gpt-5.4	1.0	0.231	0.214	0.97	[0.94, 1.00]	no	$+0.066$
gpt-5.4	0.5	0.433	0.246	1.09	[1.05, 1.13]	yes	$+0.242$
claude-sonnet-4-6	1.0	0.182	0.323	0.67	[0.66, 0.69]	yes	$+0.176$
claude-sonnet-4-6	0.5	0.421	0.373	0.75	[0.73, 0.77]	no	$+0.424$
MiniMax-M2.7	1.0	0.179	0.193	0.93	[0.89, 0.97]	yes	$+0.048$
MiniMax-M2.7	0.5	0.347	0.224	1.00	[0.96, 1.04]	no	$+0.227$

J.3. X2: Residual coverage and portfolio marginals

Results tested and hypotheses. (10); Theorem 2 under its hypothesis "the compared portfolios have common conditional values v_h, v_m and fallback cost b, and use (7) with primary bill $a(S)$", through the proof identity (39); the real-data phenomenon behind Proposition 3 (the proposition itself is a two-world construction and cannot fail on data); and Appendix E.6. Portfolio S: all members are called in parallel, the exact selecting verifier commits the best passing member, Opus is called and committed on a miss; D is the sum of all billed calls, $a(S) = \mathbb{E}[\sum_{m\in S} C_m]$, $d_m = \mathbb{E}[C_m]$ (no aggregation fee), V the committed score. Candidates are the 7 non-Opus models; contexts are the 9 dimensions and the pooled set; a candidate is usable in a context with at least 20 passes and 20 failures, and an (S, m) case is non-trivial with at least 10 tasks in each conditional cell; $|S| \in \{1, 2, 3\}$.

Prediction with plugged primitives. With S's values, $N_S = r_m(S)[v_h - v_m + \eta(S)b] - \eta(S)d_m$ and the predicted change is $N_S/(\mathbb{E}[D(S)] + d_m - r_m(S)b)$. Pooled context, $\theta = 1$, $S = \{$Qwen3-Max$\}$: adding gpt-5.4 has $N_S = -0.291$ and predicts a change of -21.3 score per dollar against a realized -22.8 $[-24.3, -21.3]$; adding qwen3.5-plus has $N_S = -0.013$, predicts -1.4, and realizes $+5.0$ $[3.5, 6.6]$ because b falls by 14% on the new failure set.

Table 3. X2 claims (kinds as in Table 1; values $\theta = 1$ / $\theta = 0.5$).

	Claim	Kind	Measurement	Verdict
a1–a3	$r_m(S) = p(S \cup \{m\}) - p(S)$; (7) equals the ratio of sums; (39)	I	max |difference| 1.1×10^{-16} / 2.4×10^{-12} / 4.6×10^{-12} over 8,336 cases	PASS
b1	$v_m(S \cup \{m\}) = v_m(S)$	IS	n = 1,457 / 1,869; median drift -13.7% / -17.5%; negative in 97% / 97%; CI excludes 0 in 84% / 85%	FAIL
b2	$b(S \cup \{m\}) = b(S)$	IS	median drift $+0.8\%$ / $+0.6\%$; 10–90% range $[-4.9\%, +4.3\%]$ / $[-7.1\%, +6.3\%]$; CI excludes 0 in 65% / 59%	FAIL (small)
b3	$v_h(S \cup \{m\}) = v_h(S)$	I / IS	exactly 0 at $\theta = 1$ (hit score is 1); at $\theta = 0.5$ median 0.00%, max |drift| 0.164, CI excludes 0 in 52%	PASS / FAIL (small)
b4	sign of N_S with S's values equals realized sign	IS	90.3% / 93.3% of all cases; 95.1% / 97.8% of realized-CI-decided; 99.0% / 99.5% of both-CI-decided (11 / 8 disagreements); optimistic disagreements 133:8 / 109:17	PASS
c1	best-quality partner $\neq$ best-coverage partner	IS + OOS	in-sample 5/47 / 3/56 contexts; held out 6/47 / 5/56 differ, gap CI > 0 in 1 / 2 (3 / 3 with the pooled context); e.g. code_generation, m_0 = glm-5: $+0.020$ $[0.006, 0.038]$	PASS (rare)
c2	$\mathbb{P}(E_m^c \cap E_i^c) > \mathbb{P}(E_m^c)\mathbb{P}(E_i^c)$	IS	136 / 168 pairs; excess > 0 in 99% / 99%, CI > 0 in 98% / 99%; median excess 0.082 / 0.090	PASS
c3	Spearman of $\mathbb{P}(E_m)$ against pair coverage	IS	median 0.93 / 0.90 (r_m against pair coverage is 1 by identity)	PASS (descriptive)
d1	sign of probing N_S predicts realized `id_test` sign	OOS	91.4% of 674 / 93.1% of 958; 96.7% / 99.0% of CI-decided	PASS
d2a	top by probing r_m vs top by $\mathbb{P}(E_m)$	OOS	differ in 9/139 / 18/194; r_m wins 6:3 / 6:12	INCONCLUSIVE
d2b	top by probing N_S vs top by $\mathbb{P}(E_m)$	OOS	differ in 58/139 / 114/194; N_S wins 58 / 114 (56 / 109 CI-significant); median gain $+5.8\%$ vs -0.3% / $+4.2\%$ vs -12.8%; oracle $+6.2\%$ / $+4.4\%$	PASS
d2c	N_S with r_m vs N_S with $(1 - p(S))\mathbb{P}(E_m)$	OOS	differ in 16/139 / 21/194; 8:8 / 13:8 (CI-significant 0:5 / 0:1)	INCONCLUSIVE
d3	probing r_m ranks realized r_m better than probing $\mathbb{P}(E_m)$	OOS	Spearman 0.900 vs 0.829, 72:36 of 139 ($p = 0.001$) / 0.900 vs 0.899, 88:62 of 194 ($p = 0.041$)	PASS / PASS (weak)
e1	$r_m(T) \leq r_m(S)$ for $S \subseteq T$	I	0 violations in 2,160 + 102,060 triples	PASS
e2	$\mathbb{P}(E_m \mid \text{all of } S \text{ fail}) > \mathbb{P}(E_m)$	IS	exceeds at the point in 0.2% / 0.1% of 1,741 / 2,156 cases, never with CI > 0; below in 99.1% / 99.5%; nested form rises with CI > 0 in 0.7% / 0.6%	NOT INSTANTIABLE

Table 4. X2 sign-rule baselines on non-trivial dimension cases (fraction of cases whose predicted sign equals the realized sign of $\eta(S \cup \{m\}) - \eta(S)$).

Rule	in-sample, $\theta = 1$ (n = 1,457)	in-sample, $\theta = 0.5$ (n = 1,869)	OOS, $\theta = 1$ (n = 674)	OOS, $\theta = 0.5$ (n = 958)
(11) with S's v_h, v_m, b	90.3%	93.3%	91.4%	93.1%
value-preserving case $r_m b - d_m$	82.7%	85.4%	81.6%	85.4%
independence proxy $(1 - p(S))\mathbb{P}(E_m)$	64.9%	66.4%	71.5%	71.2%
always negative (base rate)	70.1%	72.1%	61.9%	66.9%

Hypothesis violations measured. Substituting only $v_m(S \cup \{m\})$ into the prediction cuts the median |relative error| of the predicted change from 17.1% to 4.1% at $\theta = 1$ (10.7% to 4.2% at $\theta = 0.5$); the signed bias of (11) is $+1.6$ / $+0.9$ points of $\eta(S)$. The two binary-score dimensions (code_generation, data_science) make their $\theta = 1$ and $\theta = 0.5$ rows one experiment, so the distinct held-out c1 cases with CI > 0 are code_generation/glm-5 ($+0.020$ $[0.006, 0.038]$), code_completion at $\theta = 0.5$/claude-sonnet-4-6 ($+0.027$ $[0.003, 0.055]$), and three pooled cases, the largest $+0.032$ $[0.020, 0.042]$. Dropping the 148 zero-cost Qwen3-Max tasks changes the bug_fixing sign agreement from 81.2% to 82.5% ($\theta = 1$) and from 94.8% to 90.9% ($\theta = 0.5$) with no verdict change. The top partner by N_S and by the full predicted change differ in 1/139 and 0/194 portfolios.

J.4. X3: Attempt-then-escalate threshold and the value of routing information

Results tested and hypotheses. (15) and Proposition 5 in a state where "the primary costs $a > 0$, is sufficient with probability p, and delivers value v_h on a hit. On a miss, the fallback costs b on top of a and restores value v_b, the value of going directly to that path"; the refinement clause "a routing refinement costing k that increases p by Δp while preserving the other quantities"; (16) and Proposition 6 under "both accepted primary outputs and fallback preserve value", $0 < a < b$, and a signal "that reveals the sufficiency event exactly for cost $k \geq 0$"; and the bound $f(s) - g(s) = \min\{s(b-a), a(1-s)\}$ of Appendix C.3. States are the 9 dimensions, the 21 cells, and the pooled set; primaries Qwen3-Max and kimi-k2.5; $\lambda \in \{0.1, 1, 10, \eta_B, 100\}$ score per dollar with $\eta_B = 34.727$ on probing; $J_\lambda = \mathbb{E}[V - \lambda D]$; direct = Opus only.

Prediction with plugged primitives. Pooled, Qwen3-Max, $\theta = 1$: $p = 0.2144$, $v_h = 1$, $v_b = 0.4297$, $a = \$0.000532$, $b = \$0.01237$; at $\lambda = 1$, $W_\lambda = 0.5826$, $\lambda a / W_\lambda = 0.0009$, and the predicted $J_{\text{casc}} - J_{\text{direct}} = pW_\lambda - \lambda a = 0.124$ $[0.120, 0.129]$. For (17) in the same state $s = 0.214$ and the band is nonempty only for $k < a(b-a)/b = \$0.000509$; at $k = \$10^{-4}$ the band is $(0.0084, 0.812)$ and contains s (information), at $k = \$5 \times 10^{-4}$ it is $(0.042, 0.060)$ and excludes s (cascade).

Table 5. X3 claims (kinds as in Table 1; M = measurement without a paper prediction). "Decided" = realized CI excludes 0.

	Claim	Kind	Measurement	Verdict
C1	sign rule: attempt first iff $pW_\lambda \geq \lambda a$	OOS	9-dimension: 180 cells, PASS 148 / FAIL 12 / INCONCLUSIVE 20, decided agreement 0.925, all-cell 0.883 $[0.850, 0.906]$; 21-cell: 420 cells, 305/22/93, decided 0.933, all-cell 0.910 $[0.819, 0.926]$; baselines on decided cells: always attempt 0.763/0.639, $p \geq a/b$ 0.950/0.920, any probing hit 0.856/0.899; 17 of the 34 FAIL rows are cells with zero `id_test` hits and 22 have a prediction CI covering 0	PASS (qualified)
C2	$W_\lambda \leq 0$ implies direct strictly better	OOS	exact: 5/5 cells (one physical cell, code_completion|cruxeval at $\theta = 0.5$; Qwen3-Max, $\lambda = 1$: -0.037 $[-0.057, -0.014]$); noisy: 139 PASS / 4 FAIL / 6 INCONCLUSIVE	PASS (weak power)
C3	magnitude $J_{\text{casc}} - J_{\text{direct}} = pW_\lambda - \lambda a$	OOS	pooled Qwen3-Max $\theta = 1$, $\lambda = 1$: predicted 0.124 $[0.120, 0.129]$, measured 0.071 $[0.062, 0.080]$; kimi-k2.5 $\theta = 0.5$: 0.158 vs 0.032 $[0.025, 0.040]$; measured inside the prediction CI in 12% / 30% of cells; MAE 0.086 / 0.075; signed error $+0.086$ / $+0.070$ at $\lambda \leq 10$	FAIL
C4	miss-independent v_b: $\mathbb{E}[Y_B \mid H = 0] = \mathbb{E}[Y_B]$	IS	pooled Qwen3-Max: -0.073 $[-0.083, -0.064]$ at $\theta = 1$, -0.180 $[-0.192, -0.167]$ at $\theta = 0.5$; kimi-k2.5 -0.065 / -0.183; CI below 0 in 7/9 dimensions; $\mathbb{E}[Y_B \mid H = 1] - \mathbb{E}[Y_B] \in [+0.26, +0.35]$	FAIL
C5	refinement $\Delta p\, W_\lambda \geq \lambda k$, instantiated as a non-contingent primary swap with $k = \Delta a$	OOS	$\theta = 1$: 9-dimension 173/10/57, decided 0.945 ($n = 183$) vs never-swap 0.853; 21-cell 318/27/135, 0.922 ($n = 345$) vs 0.913; by λ 0.83/0.86/0.97/1.00/1.00 and 0.78/0.78/0.99/1.00/1.00; MAE 0.027 / 0.050	PASS (qualified); NOT INSTANTIABLE for a contingent router
C6	band algebra of Proposition 6	I	240 cell$\times k$ combinations, 0 failures	PASS
C7	information is realized-cheapest iff s is inside the band	OOS	$\theta = 1$: 202 PASS / 7 FAIL / 31 INCONCLUSIVE, decided 0.967; $\theta = 0.5$: 207/2/31, 0.990; three-way argmin 0.923 / 0.947; in-sample 1.000; baselines "band nonempty" 0.722 / 0.766 and "$k \leq \$10^{-4}$" 0.775 / 0.809	PASS
C8	a router's saving never exceeds $f(s) - g(s)$	I-like	0 of 1,656 dimension and cell rows and 0 of 72 pooled rows exceed the closed form with cell-mean a, b; largest ratio 0.952; the realized-cost bound is tautological given $C_P \leq C_B$ per task	PASS
C9	fraction of $f(s) - g(s)$ captured by each router	M	see Table 6	PASS (reported)

Table 6. X3, C9: fraction of the perfect-information value $f(s) - g(s)$ captured by each released router (pooled state, Qwen3-Max primary, $\theta = 1$, $k = 0$; Z = router chose the primary model; cascade if $Z = 1$, direct otherwise). Negative values mean the router routes more sufficient tasks to Opus than the uninformed cascade does ($\mathbb{P}(H = 1 \mid Z = 0) = 0.15 > a/b = 0.043$ for the classical routers). At $\theta = 0.5$ and with kimi-k2.5 every fraction is negative (-1.4 to -7.4).

Router	Fraction of $f(s) - g(s)$ captured
fine-tuned Qwen 0.8B	0.45 [0.37, 0.51]
fine-tuned Qwen 2B	0.50
fine-tuned Qwen 9B	0.58
fine-tuned Qwen 27B	0.59 [0.53, 0.65]
logistic regression	-1.30 [-1.70, -0.97]
TF–IDF MLP	-1.86
RouteLLM matrix factorization	-1.11
RouteLLM similarity-weighted	-1.37
DimensionBest	-1.57

Hypothesis violations measured. The homogeneous state fails through v_b: $\mathbb{E}[Y_B \mid H = 0] - \mathbb{E}[Y_B]$ as in C4, and on costs $\mathbb{E}[C_B \mid H = 1] - b = +\$0.0073$, $\mathbb{E}[C_B \mid H = 0] - b = -\$0.0021$ (Qwen3-Max, $\theta = 1$). Replacing v_b by the miss-conditional mean moves the pooled prediction from 0.126 to 0.069 against the exact 0.071; at $\lambda = \eta_B$ the value and cost biases nearly cancel for Qwen3-Max (0.196 vs 0.194) but not for kimi-k2.5 (0.141 vs 0.086). The threshold has little bite at these prices: $\lambda a/W_\lambda \leq 0.049$ in every pooled state and exceeds the probing p in 18 of 600 cell rows. The refinement swap preserves v_h exactly at $\theta = 1$ and violates it at $\theta = 0.5$ ($|\Delta v_h|$ up to 0.12, agreement 0.91 / 0.93); its one pooled FAIL is gpt-5.4 at $\lambda = 1$, predicted $+0.0024$ $[-0.0017, 0.0062]$ and measured -0.020 $[-0.027, -0.013]$. The Blackwell-optimal use of the same router signal with cell-mean costs captures 0.38 (ft-27B), 0.30 (ft-9B), and at most 0.05 for the others.

J.5. X4: Record process, retention, regret, and heavy tails

Results tested and hypotheses. (19), (23), (40), (20), (21), (22), Theorem 7 with (24), the stationary criterion and (42), and (25) under "$g_j^I > 0 > g_j^W$"; Appendix G.2; (46) and (47) and the statement that replacing E by M/β is not exact; (48); Theorem 8 under "$\nu_j = j^{-z}/\zeta(z)$, $z > 1$, writes are clean with constant probability $\omega \in (0, 1]$, and every cell has the same positive informed-minus-cold hit gain Δ"; and Lemma 12. The record model's hypothesis that "write types are independent of the past and of the current record" is realized by an explicit logger: with probability $\varepsilon = 0.25$ an epoch fields a uniformly random model, verifies its real outcome, and writes the model to the arriving cell (informative if a true hit, erroneous if a false accept; last write wins; first used at epoch $n+1$), so $\omega_j = \varepsilon(1 - \beta_{\mathrm{fr}})\,\mathrm{mean}_m P_{jm}$ and $\psi_j = \varepsilon\beta_{\mathrm{fa}}\,\mathrm{mean}_m(1 - P_{jm})$ with P_{jm} the real pass rate of model m in cell j. The informed policy fields the recorded model on non-exploration epochs; the cold policy is always-Opus or DimensionBest fitted on probing. Cells are the 21 dimension×source pairs (also the 9 dimensions); streams are 50 i.i.d. draws of 20,000 `id_test` tasks (300 for retention), stationary values on the last 5,000 epochs; $\theta \in \{1, 0.5\}$, $\beta_{\mathrm{fa}} \in \{0, 0.05, 0.1, 0.2, 0.4\}$ plus $(0.1, 0.2)$, $\delta \in \{0, 0.01, 0.05\}$; resources dollars and seconds of the fielded call; verification is free.

Predictions with plugged primitives. Cell algorithm|lcb, $\theta = 1$, $\beta_{\mathrm{fa}} = 0.2$, $\delta = 0$: $i_j^* = \nu_j\omega_j/(1 - \alpha_j) = 0.0956 \times 0.0850/(1 - 0.98872) = 0.720$ and $w_j^* = 0.280$; measured late-window record probabilities 0.716 and 0.284. (24) at $\theta = 1$, $\beta_{\mathrm{fa}} = 0$, $\delta = 0$, always-Opus cold, dollars, $N = 20{,}000$: in-sample $\sum_j \nu_j G_j[N - (1 - \alpha_j^N)/(1 - \alpha_j)] = +3357$ with $\eta_U = 42.8$ score per dollar; measured $+3402$ $[3070, 3765]$ unpaired and $+3393$ $[3108, 3707]$ paired. Out of sample (probing primitives, $\eta_U = 39.6$) the prediction is $+3484$ and the measurement $+3776$ $[3449, 4128]$ unpaired, $+3143$ $[2870, 3444]$ paired, whose CI excludes the prediction. (25) for code_completion|he, $\theta = 1$, DimensionBest, dollars: $g_j^I = 0.262$, $g_j^W = -0.090$, threshold 0.745 against $\psi_j/r_j \in [0, 0.061]$, so the cell is predicted positive. (47) for $M = 5$ at $\theta = 1$, $\beta_{\mathrm{fa}} = 0$ with $\beta = 0.0516$ writes per epoch: aggregate coverage exact 0.2775, naive $1 - (1 - \nu_j r_j)^{M/\beta}$ aggregate 0.2659, measured 0.2782 $[0.2770, 0.2793]$. (48) at $\theta = 1$, DimensionBest, dollars, $N = 20{,}000$: $10611.2 - 550.6 = 10060.6$ against a measured 9977 $[9375, 10551]$. Appendix G.2: shadow-call overhead $h = \$0.00497$ per epoch; always-Opus cold $\sum_j \nu_j G_j(\eta_0) \in [0.200, 0.230] \geq \eta_0 h = 0.187$ (memory beats the memory-free reference), DimensionBest $-0.19 < 0.57$ (memory loses).

Hypothesis violations and scope. No stated hypothesis fails here because the logger is built to satisfy (19); the informative content is which sign the criterion takes. Against DimensionBest a single accepted-model record has $g_j^I < 0$ in most cells and memory hurts on both resources; against always-Opus it helps in dollars and hurts in seconds. The negative side of (25) is exercised only out of sample in one cell (algorithm|lcb, seconds) whose in-sample $g_j^I = -0.095$ does not meet the hypothesis. Blocks (g) and (h) use real outcomes only as labels ($\Delta = 1$ by construction; fixed cell means), so they are Monte-Carlo checks of the occupancy sums and of the logging bound, not data results. The overhead h is 43% of the always-Opus and 111% of the DimensionBest memory-free bill.

Table 7. X4 claims. All blocks are identities plus sampling checks unless marked; OOS = probing primitives against an `id_test` stream.

	Claim	Measurement	Verdict
(a)	(23) record probabilities	1,106 cell×state×configuration series; 98.1% of 250-epoch bins have $\lvert z\rvert < 2$, max $\lvert z\rvert = 3.66$; late-window prediction covered in 96.1% (strict CI 87.6%)	PASS
(b)	(24) cumulative surplus	in-sample: paired CI covers 96.7% of 768 points, median normalized error 0.0074 at $N = 20{,}000$; OOS: sign 128/128, median normalized error 0.057 unpaired / 0.086 paired, paired CI covers 25% (unpaired 80%), median relative error 9.8%	PASS / PASS (OOS, qualified)
(c)	$\eta_\infty \geq \eta_U$ iff $\sum_j \nu_j G_j \geq 0$; (42)	in-sample and OOS: TP 32 / TN 96 / FP 0 / FN 0 over 128 configurations (four distinct sign cases: cold × resource); magnitude CI covers the prediction in 99.2%, median relative error 1.5%; always-Opus, dollars: η from 42.9 to 72–83; 9-cell variant TP 4 / TN 12	PASS / PASS
(d)	(25) where $g_j^I > 0 > g_j^W$	hypothesis met in 144 of 2,604 in-sample cell configurations (3 cells) and 128 OOS (5 cells); in-sample 84 conclusive, all true positives; OOS TP 62 / TN 14 / FP 3 / FN 0 (0.962), the 3 FP in one cell with $g_j^I = +0.0003$	PASS (one-sided) / PASS
(e)	(46), (47); $E \to M/\beta$ not exact	300-stream replay, 462 points: TTL covered 96.8% (strict 84.8%), FIFO 94.8% (strict 80.7%); of 84 resolvable $M \leq 20$ points the measurement is closer to the exact formula in 97.6%, naive rejected 97.6%, exact 3.6%; aggregate $M \leq 20$: naive rejected 12/12 (mean $\lvert z\rvert$ 16.6), exact mean $\lvert z\rvert$ 0.67	PASS / PASS / PASS
(f)	(48)	in-sample CI covers 98.3% of 288 points, median relative error 0.34%; OOS 3.1%, covered 91.7%; transient term negative in 26/48 configurations	PASS / PASS
–	Appendix G.2 identity and threshold	$J_U(\eta_0) + \eta_0 h = 0$ to 10^{-16}; identity CI covers 8/8 (median relative error 0.9%); threshold sign 8/8 (two sign cases: always-Opus $+1.6$ to $+4.7$, DimensionBest -72 to -74 score per dollar)	PASS / PASS
(g)	Theorem 8 exponents (semi-synthetic Zipf over 234 real tasks, $J = 20{,}000$, $\omega = 0.5$, $\Delta = 1$)	TTL slopes $-0.349/-0.504/-0.672$ for $z = 1.5/2/3$ (target $-0.333/-0.500/-0.667$), $E \in [200, 2000]$; cumulative $0.649/0.503/0.346$ (target $0.667/0.500/0.333$), $N \in [2000, 20000]$; seed CI covers the exact finite-J sums 48/48; a control violating common ω, Δ gives $-0.27/-0.80/-0.59$	PASS / PASS
(h)	Lemma 12 bound $(1 - e_{m^*})^v$	violations (measured $>$ bound $+2.5$ SE) at most 0.24% of 2,100 (cell, v) points per configuration, e_{m^*} of 0.02, 0.05, 0.1, or 0.2, 4,000 replicates; measured/bound 0.57–0.83; convergence to the best arm in the logger's support at $v = 200$ in 21/21 cells	PASS

J.6. X5: Two-state hit dynamics, hedging, and the barrier round

Results tested and hypotheses. Proposition 10 under "$(H_n)_{n\geq 1}$ is a homogeneous Markov chain with transition probabilities p_h and p_m, where $0 < p_m \leq 1$, $0 \leq p_h < 1$, and $(p_h, p_m) \neq (0,1)$" and "$\mathbb{E}[V_n \mid \mathcal{F}_{n-1} \vee \sigma(H_n)] = v_{H_n}$, $\mathbb{E}[D_n \mid \mathcal{F}_{n-1} \vee \sigma(H_n)] = d_{H_n}$", together with the caveat of Appendix B that a stationary binary process satisfies flow balance without the Markov property; Appendix D.2 under "deterministic completion times satisfy $0 < \ell_p \leq \ell_b$" with "hit value v_h and matched miss/reference value $\bar{v} > 0$", conditions (33) and (34) with cancellation prefix charge $r \in \{0, 0.1b, b\}$; Proposition 13 under "$L > 0$ the deterministic primary-round latency" and "$V_\mathrm{s} > V_\mathrm{f} > 0$, $\Delta_\mathrm{s} > \Delta_\mathrm{f} \geq 0$, and $0 < \pi_\mathrm{f} < \pi_\mathrm{s} < 1$". Part (a) uses a random permutation of the `id_test` tasks (the benchmark records no arrival order). Part (b) uses four constructions with primitives from probing and 10 streams of 10,000 epochs plus 300,000 conditioned chains: (1) after a miss the next task is drawn from the hard tercile of a random dimension, after a hit from the pool (exactly Markov); (2) the same with the recovery dimension equal to the current one with probability 0.5; (3) a hidden three-regime stream (regime = difficulty tercile, persistence 0.8; stationary, non-Markov); (4) an exact two-state chain with the (p_h, p_m) of (1). Literal same-dimension recovery is absorbing ($p_m = 0$) and NOT INSTANTIABLE. Part (c) uses all 7 primaries × 2 thresholds on latency-valid tasks with $\ell_\kappa = 0$. Part (d) forms rounds of fan-out $b \in \{1, \ldots, 64\}$ from 64 member slots (Qwen3-Max primary, 800 rounds after burn-in, residue probability $\rho \in \{0.3, 1\}$); the slow policy re-runs Opus in parallel on missed members, the fast policy commits the cheap output ($\Delta_\mathrm{f} = 0$) or a gpt-5.4 repair.

Predictions with plugged primitives. Qwen3-Max, $\theta = 1$, shuffled order: $p_h = 0.195$, $p_m = 0.222$, CI of the difference $[-0.022, +0.018]$; in the file's dimension-blocked order $p_h = 0.611$, $p_m = 0.107$. Hedging, Qwen3-Max, $\theta = 1$: the plug-in margin $p(\ell_b v_h - \bar{v}\ell_p)$ from probing is $+0.179$ s (hedging predicted to beat $\bar{v}/\ell_b$), while the realized $\eta^\ell_{\text{hedge}} - \bar{v}/\ell_b$ on `id_test` is -0.0294 $[-0.0322, -0.0270]$ per second. Dollars, same cascade: $\eta^c_{\text{hedge}} - \eta_B = +25.5$ $[21.9, 28.9]$ per dollar at $r = 0$ and $+3.9$ $[3.2, 4.5]$ at $r = b$. Barrier, commit policy, $\theta = 1$, $\rho = 0.3$: $\Xi = 0.280$, $b^* = \log \Xi / \log \pi_\mathrm{s} = 0.74 < 1$, so the fast policy is predicted better at every fan-out; the empirical regime is fast-everywhere in 91% of bootstrap replicates.

Table 8. X5: mean latency of each model on latency-valid `id_test` tasks. Opus is the fastest model, so the hedging hypothesis $\ell_p \leq \ell_b$ fails for every primary; the per-task probability that the primary finishes before Opus is 0.00–0.52.

Model	Mean latency (s)
claude-opus-4-6 (reference)	7.8
claude-sonnet-4-6	8.2
gpt-5.4	8.2
kimi-k2.5	12.4
Qwen3-Max	16.3
MiniMax-M2.7	25.7
qwen3.5-plus	41.2
glm-5	44.9

Hypothesis violations measured. Both hedging hypotheses fail on every cascade (C1, C2), so the plug-in FAILs of C4 and C5 are applicability failures of a conditional statement, not counterexamples; the state-matched margins equal $\mathbb{E}[V]\ell_b - \bar{v}\mathbb{E}[\text{latency}]$ (in-sample identity gap 7×10^{-15}), so their PASS only shows that the sign of Theorem 1 transfers across the split. The probing→`id_test` shift is systematic: Opus cost -5.8% (Welch $p = 0.027$), latency -5.7% ($p = 0.015$), score $+2.0\%$ ($p = 0.36$); dimension and source composition are balanced (χ^2 $p = 0.56$ / 0.68). Part (c) dollar results for glm-5, qwen3.5-plus, and MiniMax-M2.7 drop 5–15% of tasks with unlogged latency.

Table 9. X5 claims, part 1: i.i.d. stream (A) and constructed streams (B). CC = construction check; I = identity; IS = in-sample hypothesis; OOS = out-of-sample; SS = semi-synthetic construction. Verdicts after audit.

	Claim	Kind	Measurement	Verdict
A1	$p_h = p_m$ on the shuffled stream	CC	9/9 CIs cover 0, max $\lvert p_h - p_m\rvert = 0.026$; in the file's own order $p_h - p_m \in [+0.27, +0.50]$ (permutation-null z 27–46)	PASS by construction
A2	$\pi = p_m/(1 - p_h + p_m)$ equals the hit frequency; ratio identity	I	max $\lvert\pi - \text{freq}\rvert = 7.9 \times 10^{-5}$ ($1/N = 10^{-4}$); ratio residual 6.7×10^{-16}	PASS
A3	first-half primitives predict second-half ratio of sums (dollars, sequential s, hedged s)	OOS	coverage 0.96 / 0.94 / 0.91, mean |rel. err.| 3.3% / 2.2% / 2.1%, signed $+0.6\%$; an alternative draw of the same 10 splits gives 0.91 / 0.88 / 0.88 and -1.1%	PASS at the stored seed, FAIL at the alternative draw
B0	literal same-dimension hard-tercile recovery has $p_m > 0$	IS	$p_m = 0$ in 4/4 cases (absorbing)	NOT INSTANTIABLE
B1	Markov property: $\mathbb{P}(1 \mid 00) - \mathbb{P}(1 \mid 10)$	SS	(1) $+0.002$, 0/4 rejected; (2) $+0.002$, 0/4; (3) -0.072 to -0.140, rejected 4/4; (4) $+0.002$, 0/4	(1),(4) PASS; (2) INCONCLUSIVE; (3) PASS as the caveat instance
B2	π from probing predicts the realized hit frequency	OOS	$\lvert z\rvert \le 1.74$ in 16/16; relative error (1) $+3.7$ to $+7.0\%$, (2) -0.1 to $+4.0\%$, (3) -5.3 to $+0.6\%$, (4) 0.0 to $+0.7\%$; (4) shares its parameters with the stream	PASS ((4) vacuous; (3) INCONCLUSIVE under the first z rule)
B3	geometric transient $\mathbb{P}(H_n = 1 \mid H_1 = 0) = \pi(1 - \xi^{n-1})$, $n = 3..9$	SS	two-state max $\lvert z\rvert$: (1) 2.1, 2.1, 1.5, 1.9; (4) 1.7, 1.7, 1.4, 2.2; (2) ≤ 1.8; (3) 17.4, 44.4, 15.1, 41.6 (gap 6.5–12% of π) while the exact 6-state chain has ≤ 2.1	(1),(4) PASS; (2) INCONCLUSIVE; (3) PASS: transient fails as the caveat states
B4	$\mathbb{E}[V_n \mid \mathcal{F}_{n-1} \vee \sigma(H_n)] = v_{H_n}$	IS	$\mathbb{E}[V \mid \text{miss, prev. miss}] - \mathbb{E}[V \mid \text{miss, prev. hit}]$: (1)–(3) -0.11 to -0.25, CI excludes 0 in 12/12; (4) |gap| ≤ 0.005	FAIL (1)–(3); PASS (4)
B5	long-run ratio from probing stationary means	OOS	relative error (1) -1.8 to $+1.7\%$; (2) -4.7 to $+1.3\%$; (3) -8.8 to -4.7%, z_{full} -2.5 to -1.2; (4) -7.1 to -4.8%, $z_{\text{full}} \le 1.6$; bias = probing→`id_test` shift	(1),(2),(4) PASS; (3) INCONCLUSIVE

Table 10. X5 claims, continued: hedging (C) and barrier rounds (D). Kinds as in Table 9.

	Claim	Kind	Measurement	Verdict
C1	$0 < \ell_p \leq \ell_b$	IS	0/14 (Table 8)	FAIL 14/14
C2	$v_m = \bar{v}$	IS	$v_m \in [0.15, 0.37]$ against $\bar{v} = 0.44$, CI excludes 0 in 14/14	FAIL 14/14
C3	mean-latency formulas $\ell_p + (1-p)\ell_b$ and $p\ell_p + (1-p)\ell_b$	OOS	sequential plug-in covered 9/14 (error $+0.8$ to $+9.9\%$); hedged plug-in 1/14 (-5 to -70%); conditional-mean forms 11/14	FAIL (90% rule)
C4	sign of (33)	OOS	plug-in 6 PASS / 8 FAIL (4 primaries predict a win at both θ and lose); state-matched 14/14; realized $\eta^{\ell}_{\mathrm{hedge}} < \bar{v}/\ell_b$ in 14/14	FAIL (plug-in; hypotheses violated) / PASS (state-matched, a Theorem 1 transfer check)
C5	sign of (34), $r \in \{0, 0.1b, b\}$	OOS	plug-in 10/3/1, 11/3/0, 10/2/2 (PASS/FAIL/INCONCLUSIVE; failures at $\lvert\text{margin}\rvert < \$0.001$); state-matched 13/0/1, 14/0/0, 12/0/2	FAIL (plug-in) / PASS (state-matched)
C6	hedged latency below sequential	I	saving 6.3% (claude-sonnet-4-6, $\theta = 0.5$) to 38.3% (glm-5, $\theta = 1$)	PASS
D1	$V_{\mathrm{s}} > V_{\mathrm{f}} > 0$, $\Delta_{\mathrm{s}} > \Delta_{\mathrm{f}} \geq 0$, $\pi_{\mathrm{f}} < \pi_{\mathrm{s}}$	IS	value and latency orderings 8/8 (V_{s} 0.24–0.44 vs V_{f} 0.17–0.40; Δ_{s} 6.8–7.1 s vs Δ_{f} 0 or 6.0–6.5 s); $\pi_{\mathrm{f}} < \pi_{\mathrm{s}}$ at the point 7/8, CI excluding 0 in 4/8	PARTIAL
D1b	deterministic round latency L independent of b	IS	primary-latency CV 0.93; round latency at $b = 64$ is 4.2–4.5 $\times$ L (15.9 s to 67 s)	FAIL
D2	crossing exists iff $V_{\mathrm{s}}/(L + \Delta_{\mathrm{s}}) < V_{\mathrm{f}}/(L + \Delta_{\mathrm{f}})$	OOS	6 PASS (gpt-5.4 repair 4/4: no crossing predicted, slow better in $\geq 98.8\%$ of replicates; commit $\theta = 1$, $\rho = 0.3$: $b^* = 0.74$, fast everywhere in 91%), 2 INCONCLUSIVE, 0 FAIL; regime agreement on the grid 5/7 identifiable pairs for both instantiations	PASS 6 / INCONCLUSIVE 2
D3	at most one crossing	OOS	robust (2 SE) sign changes 0 in 8/8 pairs (no power); raw count ≤ 1 in 98.8–100% of replicates	PASS
D4	$b^* = \log \Xi / \log \pi_{\mathrm{s}}$ inside the empirical CI	OOS	$\theta = 0.5, \rho = 0.3$: empirical 1.64 $[1.09, 4.94]$, formula 1.05; $\theta = 1, \rho = 1$: 5.84 $[2.83, 52.6]$, formula 1.10; $\theta = 0.5, \rho = 1$: 38.7 $[9.2, 60.5]$, formula undefined ($\Xi = -0.247$); an order-statistic instantiation of $L(b)$ is inside in 2/3	FAIL 3/3 (deterministic-L hypothesis violated)

J.7. X6: Coverage allocation across positions

Results tested and hypotheses. Proposition 11 under "K independent positions or task families have increasing, strictly concave differentiable coverage curves $p_k(F_k)$, where $F_k \geq 0$ is a continuous allocation and $\gamma_k > 0$ its unit price" and, for (36), "$1 - p_k(F) = d_k(1+F)^{-r}$ with $d_k \in (0, 1]$ and common $r > 0$"; the active-set procedure (37); (38); Appendix E.5 with weights $w_k = \nu_k(v_{h,k} - v_{m,k} + \lambda b_k)$ under "fixed hit/miss values, and fallback cost", the penalized KKT condition $w_k p_k'(F_k) = \lambda\rho_k + \Lambda\gamma_k$, and the statement that "the budget need not bind in this penalized problem"; and the remark that Theorem 8 motivates $r = 1 - 1/z$ only asymptotically. Positions are the 9 dimensions (and the 21 cells); $F_k \in \{0, \ldots, 7\}$ is the number of prepared lower-priced models in a verified cascade whose unit order is fixed on probing (price order, or a coverage-greedy order that is concave by construction); $p_k(F)$ is the probing probability that at least one prepared model has score $\geq \theta$; $w_k = \nu_k$ and $\gamma_k = 1$ (unit prices) or the linearized dollar price. Allocators use probing only (exact integer dynamic program, closed form with active set, marginal equalization with per-position fits, exponential closed form, discrete greedy, and uniform averaged over its K rotations) and are scored on `id_test` with 1,000 paired draws; the exact `id_test` optimum is reported only as a ceiling.

Prediction with plugged primitives. Here $p_k(0) = 0$, so the paper's $d_k = 1 - p_k(0) = 1$ for every position. With $w_k = \nu_k \in [0.108, 0.114]$, $\gamma_k = 1$, and the literal common fit $r = 0.261$, $g_k = (w_k d_k/\gamma_k)^{1/(1+r)}$ varies by less than 5% across positions and $F_k^\star = [\mu g_k - 1]_+$ is uniform up to rounding, so the dimension-level comparison with uniform is uninformative by construction. A free-intercept variant (d_k fitted, $r = 0.074$ $[0.071, 0.077]$) is a different model that routes budget to the positions with $d_k = 1$, which are the never-hit ones; its numbers are reported separately. Under the penalized problem with $\lambda = \eta_B = 34.73$ per dollar, spending stops at 17 of 63 slots at $\theta = 1$ for both the penalized KKT allocation and the exact J_λ optimum.

Table 11. X6 fits of the coverage curves on probing (dimensions, price order). RMSE in coverage units; own-r_k CIs from 500 probing resamples; values at 0.001 are search-box bounds.

Fit	$\theta = 1$	$\theta = 0.5$
common r, free d_k: r (RMSE)	0.074 $[0.071, 0.077]$ (0.157)	0.199 $[0.186, 0.211]$ (0.200)
common r, literal $d_k = 1$: r (RMSE)	0.261 $[0.253, 0.269]$ (0.274)	0.493 $[0.481, 0.506]$ (0.283)
own r_k, free d_k: RMSE	0.043	0.067
own r_k: range	0.001 (bound) to 1.601 $[1.416, 1.836]$ (test_generation); bug_fixing 0.137 $[0.119, 0.158]$, algorithm 0.608 $[0.565, 0.655]$	0.003 to 1.603
RMSE(common) − RMSE(own), 95% CI	$[0.110, 0.118]$	$[0.128, 0.136]$
exponential, own ϑ_k: RMSE	0.051	0.082

Realized cascade and sensitivities. The exact J_λ allocation at $B = 27$, $\theta = 1$, evaluated on `id_test`, scores 0.554 for \$21.04 (η = 76.9 per dollar, 21.8 s mean latency) against always-Opus 0.438 for \$34.02 (37.6 per dollar, 7.7 s); this is an oracle-verifier ceiling because every hit commits an output with score $\geq \theta$ by construction. The held-out and in-sample exact optima differ by -0.0006 $[-0.0011, -0.0000]$, so the curves transfer. Dropping the 148 zero-cost Qwen3-Max tasks changes no gap by more than 0.012 (literal closed form vs optimum -0.0427, KKT vs optimum -0.0103, KKT vs uniform $+0.0285$, common r 0.075). The uniform baseline of the original run gave its remainder slots to the alphabetically first positions and was inflated by $+0.0024$ / $+0.0041$; the rotation-averaged baseline is used here.

Table 12. X6 claims, part 1 (kinds as in Table 1; SS = semi-synthetic). Held-out gaps are differences of mean `id_test` coverage value averaged over budgets B, dimensions, unit prices, price order, $\theta = 1$ unless stated; brackets are 95% paired CIs.

	Claim	Kind	Measurement	Verdict
C1	p_k increasing and strictly concave	IS	marginal gain increases by more than 0.02 (about one SE) at 3/9 positions at each θ: algorithm $+0.228$, code_completion $+0.143$, bug_fixing $+0.055$ ($\theta = 1$); code_refactoring $+0.253$, algorithm $+0.200$, multi_language $+0.030$ ($\theta = 0.5$); code_understanding has $p_k \equiv 0$ at $\theta = 1$; coverage order is concave on probing by construction but non-concave on `id_test` at 3/9 and 5/9	FAIL (price order) / PARTIAL (coverage order)
C2	common saturation exponent r	IS	Table 11	FAIL
C3	(36) reaches the held-out optimum	OOS	literal $d_k = 1$: -0.0385 $[-0.0428, -0.0347]$ (-11.4%); $\theta = 0.5$ -0.0393 $[-0.0429, -0.0356]$; coverage order -0.0157; cells -0.0367; dollar -0.0212 / -0.0139. Free-intercept variant: -0.0912 $[-0.0979, -0.0852]$ (-27%), of which the integer projection accounts for $+0.0022$	FAIL
C4	(36) beats uniform	OOS	literal: dimensions $+0.0021$ $[+0.0010, +0.0033]$ / -0.0010 $[-0.0022, +0.0003]$ (uniform by construction); cells, unit prices $+0.0246$ $[+0.0195, +0.0296]$ / $+0.0421$ $[+0.0370, +0.0469]$; cells, dollar prices $+0.0008$, $+0.0033$, -0.0198, -0.0133. Free-intercept variant -0.0506 $[-0.0549, -0.0469]$	uninformative (dimensions) / PARTIAL (cells)
C5	KKT equal price-normalized marginals with per-position fits	OOS	vs optimum -0.0104 $[-0.0121, -0.0087]$ (-3.1%), $\theta = 0.5$ -0.0139; vs uniform $+0.0302$ $[+0.0272, +0.0336]$ $(+10.2\%)$, $\theta = 0.5$ $+0.0244$ $[+0.0219, +0.0267]$; cells $+0.0463$ / $+0.0565$. Exponential form: -0.0211 (-6.2%) / -0.0319 vs optimum, below uniform under coverage order. Greedy: -0.0076 under price order; $+0.0001$ $[0.0000, +0.0002]$ under coverage order (exact result for concave unit-price problems)	PASS (heuristic outside the hypotheses); exponential FAIL

Table 13. X6 claims, continued (kinds as in Table 12).

	Claim	Kind	Measurement	Verdict
C6	closed forms satisfy KKT; (37); exponential multiplier	I	marginal spread $\leq 5.8 \times 10^{-16}$; active set equals the level-set search at 100% of budgets in 16/16 configurations; exponential $\log \Lambda$ vs bisection 1.9×10^{-12}	PASS
C7	$1 + F_k^\star \propto (w_k d_k / \gamma_k)^{1/(1+r)}$ at the empirical optimum	IS	with $d_k = 1$ the regressor $\log \nu_k$ has range 0.05 in dimensions (no leverage); cells slope $+0.29$, Spearman $+0.70$ vs predicted 0.78	not an independent test (redundant with C2)
C8	Appendix E.5: fixed values; decomposition; budget need not bind	IS / OOS	$W_\lambda > 0$ at 8/9 ($\theta = 1$) and 9/9 ($\theta = 0.5$) positions; $v_{m,k}$ moves by up to 0.366 ($\theta = 1$) and 0.748 ($\theta = 0.5$) with F_k, pooled $v_m - v_b$ to -0.335 / -0.601; decomposition R^2 0.889 / 0.352; penalized KKT vs exact J_λ optimum -0.0082 $[-0.0120, -0.0045]$ / -0.0261 $[-0.0320, -0.0208]$; penalized vs hard-budget coverage $+0.1184$ $[+0.1087, +0.1286]$; vs uniform $+0.1356$ $[+0.1283, +0.1427]$; hard budget at $B = 45$ has $J_\lambda = 0.063$ vs 0.304	FAIL (fixed values) / PASS (decomposition at $\theta = 1$ only) / PASS (budget need not bind)
C9	$r = 1 - 1/z$ asymptotic, not at finite capacity	SS	fine family (2,919 cells), $\omega = 1$: slope 0.370/0.503/0.668 for $z = 1.5/2/3$ on $E \in [64, 1023]$, 0.443/0.510/0.667 on $[1024, 16383]$; on $F \in [0, 7]$: 0.353/0.681/0.827 (least squares in p), 0.411/0.640/0.694 (log-log); coarse 21-cell family has no power law; heterogeneous real Δ_j (mean 0.162) leaves the $z = 2$ exponent at 0.501; simulation vs (46) max $\lvert \Delta p \rvert = 0.0015$	PASS
C10	noisy verifier $\beta_{\mathrm{fa}} = \beta_{\mathrm{fr}} = 0.1$	SS	allocating from declared hits moves budget to never-hit positions: true pass 0.297 vs 0.311 at $B = 27$, 0.319 vs 0.333 at $B = 45$ ($\theta = 1$); 0.494 vs 0.509 ($\theta = 0.5$); no CIs	reported (robustness check, not a hypothesis of the proposition)

J.8. Statistical caveats

Dependence across configurations. Counts of configurations overstate independent evidence. In X4 the 128 stationary-sign configurations contain four distinct sign cases (cold policy $\times$ resource) and the overhead threshold's 8/8 is two cases; the four retention configurations share one replay stream, so per-cell deviations are correlated. In X3 the exact $W_\lambda \leq 0$ branch rests on one physical cell counted five times across primaries and λ, and the 4 noisy FAILs of C2 are one cell of 32 tasks. In X2 two dimensions have binary scores, so their $\theta = 1$ and $\theta = 0.5$ rows are one experiment, and the in-sample c1 fractions are post-selection. In X1 the noisy-verifier flips are drawn once per task and held fixed across bootstrap rows, so those CIs omit coin-flip variance; the same holds in X3 and X5. In X5 the A3 verdict flips from PASS to FAIL under a different draw of the same ten random half-splits (coverage 0.96/0.94/0.91 against 0.91/0.88/0.88; signed error $+0.6\%$ against -1.1%), so the $\pm 1\%$ / 90% rule sits at the sampling noise of 3,500-task halves. Probing and `id_test` are random splits of the same pools; every out-of-sample test measures split stability, not distribution shift, and X5 finds a 6% shift in Opus cost and latency between the splits.

Pass criteria revised after seeing data. X1: the C4 rejection threshold was relaxed from 90% to more than 50% of configurations after 24/28 rejections were seen; at the pre-registered rule C4 reads INCONCLUSIVE, and both readings are stated. X3: C5 was first declared NOT INSTANTIABLE with a router-contingent construction that Appendix C excludes; the audit added the non-contingent primary swap, whose PASS is qualified; the closed-form Blackwell comparison and the C7 baselines were also added at audit. X4: after the first full run the record-probability test received an SE floor and a 0.005 tolerance (first run: 81% of bins, FAIL, caused by zero seed SEs at coverage exactly 1 or 0), the retention block moved to a dedicated 300-stream replay with FIFO measured on full-FIFO epochs (first run: FIFO 0.83 FAIL, inexactness INCONCLUSIVE), and the Zipf family grew from 4,000 to 20,000 cells (first-run $z = 1.5$ deviations 0.033/0.051); a NaN bookkeeping bug that counted 18 undefined FIFO points as failures was fixed at audit (FIFO coverage 90.7% to 94.8%). X5: B2 and B5 are reported under both the first z rule and z_{full} (adding `id_test` pool variance), D2 received a per-pair INCONCLUSIVE rule, D2b is reported per pair as well as under the all-pairs rule, and D1 is reported PARTIAL rather than FAIL; all four changes were made after seeing results and are disclosed. X6: the uniform baseline's tie-break was corrected at audit (no verdict flips); the paper-literal $d_k = 1$ closed form replaced the free-intercept variant as the headline after audit. X2: no rule changed; a mis-described diagnostic on excluded trivial cases was corrected.

Alternative readings. X1 reports the floor under both $b = \bar{c}$ (stated) and $b = \bar{c}_m$ (realized fallback bill): sign agreement 19/28 against 24/28, value error larger under the latter; C5 and C8 are FAIL under both. X6 reports both the literal and the free-intercept closed form. X5 reports plug-in and state-matched forms of the hedging conditions. Verdicts are stated for the paper's wording; alternatives are sensitivities.

J.9. Scripts, data, and seeds

Data.

- Outcome matrix: `data/coderouterbench/id_results_long.csv` of the released Agent-as-a-Router repository.
- Price list: `data/coderouterbench/models.json` of the same repository.
- Released router decisions (nine files, used by X1, X3, and X4): `data/id/voter_decisions/` of the same repository.

Scripts. Each experiment is one self-contained script (`numpy`, `pandas`, `scipy`, `matplotlib`) that writes `results.json`, regenerates its `README.md` and vector figures, and asserts the data cross-checks. All live under `experiments/`:

- `X1_surplus_cascade/run.py` (10 s); `X2_residual_coverage/run.py` (7 s); `X3_threshold_voi/run.py` (14 s).
- `X4_memory_tail/run.py` (47 s); `X5_markov_latency_barrier/run.py` (139 s); `X6_allocation/run.py` (33 s).
- Originals: `run_orig.py`, `results_orig.json`, `README_orig.md`, `figures_orig/` in each directory; X5 also keeps `results_audit_altseed.json`; X4 keeps its independent checks under `audit/`.

- Per-cell tables written by X3: `a_threshold_cells.csv` (C1), `b2_refinement_swap_cells.csv` (C5), `c_voi_band_cells.csv` (C7), and `d_blackwell_cells.csv` (C8, C9).
- Summary written by X6: `summary_table.csv`.
- Figure 3: `make_figure.py` in the same directory, drawn from the audited `results.json` of X1, X3, and X4 only.

Seeds and draws. Every script sets seed 20260911 for its task-resampling bootstrap and constructions. X4 additionally uses seed 20260911 for the 50 streams, 777 for the task-cluster bootstrap, 99 for the seed bootstrap, 555 for the dedicated 300-stream retention replay, 4242 for the Zipf construction, and 31337 for the arm-reveal block; X5's audit diagnostic uses seed $20260911 + 99$; X6 uses 1,000 evaluation and 500 fit resamples. Bootstrap draws: X1 1,000 paired; X2 500; X3 1,000 per cell; X4 500 (task clusters and seeds); X5 500 with 100-epoch blocks for streams; X6 1,000 and 500. Re-running each original script reproduced its stored `results.json` with 0 numeric mismatches.